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AS A PANACEA FOR SUSTAINABLE  
DEVELOPMENT IN NIGERIA.

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BOOK OF ABSTRACTS



## Application of Magnetic Survey Methods for Environmental Monitoring and Sustainable Development in Nigeria

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### Abstract

*Magnetic surveying, a core geophysical technique, has proven to be an essential non-invasive tool for subsurface investigations and environmental monitoring. This paper explores the application of magnetic methods in addressing some of Nigeria's key developmental and environmental challenges, such as waste site characterization, groundwater protection, mineral resource exploration, and infrastructure planning. By measuring variations in the Earth's magnetic field, magnetic surveys can help identify buried metallic objects, geological structures, and pollution plumes without disturbing the environment. The technique's portability, cost-effectiveness, and efficiency make it highly suitable for environmental baseline studies, land-use planning, and monitoring of anthropogenic impacts, especially in urban and semi-urban areas. This paper highlights case studies and practical applications of magnetic surveys in Nigeria and discusses how this geophysical tool supports interdisciplinary efforts toward sustainable development. It also emphasizes the potential for integrating magnetic data with digital technologies like GIS and remote sensing to improve decision-making in environmental protection and resource management. The study concludes by calling for increased awareness, funding, and training in magnetic geophysical methods to enhance environmental resilience and sustainable growth in Nigeria.*



## Artificial Intelligence and its Role in Nation Development

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### Abstract

*A machine with artificial general intelligence should be able to solve a wide variety of problems with breadth and versatility similar to human intelligence. Artificial Intelligence (AI) has a wide range of applications across various industries, transforming the way we live, work, and interact. These applications showcase how AI is revolutionizing various industries, enhancing decision-making, efficiency, and innovation. However, it's important to note that AI systems also raise ethical and privacy concerns that need to be carefully addressed as technology continues to advance. These paper identify the various ways in which artificial intelligence could be applied in fields such as medical diagnosis, drug delivery, finance and autonomous vehicles.*



## Physical Sciences: Current Trends and Modern Perspectives

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### Abstract

*Physical sciences have played and continue to play an important part in the advancement and development of nations and communities in the contemporary period. Physical sciences are currently promoting new businesses and innovating information about recent breakthroughs in physics, both basic and technical. Scientists and engineers have been putting their efforts for years to prepare data storage more concise, speedy, reputable, less power consumption and accessible. Because of changes in the world new technology has seek remarkable revolution in it and one of them is flash memory which is known as its renowned application the solid-state disk or solid state derive (SSD). The present work identifies the recent development in the field of physical sciences which ranges from characterization methods for photocatalysts, synthesis of metal oxides and monte carlo methods.*



## Development of A Python Program to Estimate the Parameters of Regression Analysis

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### Abstract

*This research development of a python program to Estimate the parameters of simple linear regression was aimed to developed a program that will serve as teaching aids in statistical classes. Many students find it very difficult to understand some basic statistical application to their field in which simple Linear regression analysis is one of the commonest topics that applied to various fields in higher institutions of learning, therefore this program use a foundational statistical approach for modeling the relationship between two variables. The program allows users to input datasets and computes the best-fit linear regression line by minimizing the sum of the squares of the residuals (errors). It is designed to offer a simple, efficient and interpretable solution for analyzing linear relationships between independent and dependent variables. Python was chosen for its simplicity, readability, and strong support for scientific computing through libraries such as tkinter, NumPy and Matplotlib. The results of the regression analysis, including the slope, intercept and visual plot, provide valuable insights for data-driven decision makings in various fields such as humanities, economics, engineering and the social science. This program also can be use as teaching aids in higher institution of learning.*

**Keywords:** Program, Python, Parameters and Teaching Aids



## Ameliorative Effects of Ethanol Extract of *Azadirachta indica* (Neem) Leaves on Butyl Acetate Induced Hepatotoxicity and Clastogenicity in Male Albino Rats

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### Abstract

*Medicinal plants, as documented in indigenous pharmacopeias, possess significant therapeutic properties. These plants are used to treat various health conditions and have gained widespread acceptance in both developing and developed countries. This study investigates the protective effect of ethanol extract from *Azadirachta indica* leaves against butyl acetate-induced hepatic toxicity and clastogenicity in male albino rats. Twenty male albino rats were divided into four groups of five each: Group A received distilled water only (negative control), Group B received butyl acetate only (positive control), Group C received ethanol extract of *A. indica* and Group D received butyl acetate along with the crude ethanol extract of *A. indica*. Phytochemical screening, as well as alkaline phosphatase (ALP) levels and micronucleus assays, were examined using standard methods. The phytochemical analysis revealed the presence of saponins (89.41 mg/100g), terpenoids (42.14 mg/100g), tannins (17.41 mg/100g), alkaloids (18.27 mg/100g), phenols (0.15 mg/100g), flavonoids (0.36 mg/100g), and reducing sugars (13.21 mg/100g). The highest ALP level was recorded in the butyl acetate only ( $432.21 \pm 19.4 \mu\text{l}$ ), while the lowest was observed in the ethanol extract of *Azadirachta indica* leaves ( $199.13 \pm 1.83 \mu\text{l}$ ),  $p < 0.05$ . The micronucleus assay indicated a significantly high induction of micronucleated polychromatic erythrocytes (mPCEs) in the butyl acetate only ( $17.4 \pm 1.5 \text{ mPCEs/1000 cells}$ ), while the lowest was recorded in the distilled water ( $0.6 \pm 0.8 \text{ mPCEs/1000 cells}$ ),  $p < 0.05$ . In conclusion, the findings suggest that ethanol extract of *A. indica* mitigates butyl acetate induced hepatic injury and chromosomal damage, potentially due to its rich phytochemical composition.*

**Keywords:** *Azadirachta indica*; clastogenicity, hepatotoxicity.





## Heavy Metal Pollution and Microbial Load In Top Soil from Selected Dumpsites in Abeokuta, Ogun State and Its Public Health Implication

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### Abstract

*Dumpsites are traditional waste disposal areas harbouring a number of pollutants of environment and health concern. This study evaluates the concentration of heavy metals (HMs) and microbial loads in topsoil from selected dumpsites (Saje, Akomoje, Mokola, and Obantoko) in Abeokuta, Ogun State, Nigeria. Samples were collected, processed, and digested using Aqua regia before analysis. The Level of HM were estimated using AAS, while microbial analysis was conducted following standard methods. Pollution intensity was measured using the contamination factor (CF). The results revealed that the highest levels of HMs were as follows: lead (Pb) was highest in Mokola (9.75 mg/kg), cadmium (Cd) in Obantoko (7.24 mg/kg), chromium (Cr) in Saje (5.76 mg/kg), and arsenic (As) in Akomoje (7.37 mg/kg). The levels of Cd and Cr in Saje, Mokola, and Obantoko exceeded the WHO thresholds (0.5 and 0.2 mg/kg), respectively. Additionally, As levels in all sites surpassed the WHO threshold (1.5 mg/kg). The highest CF was recorded for As in Saje (0.55), Akomoje (0.57) and Obantoko (0.50), while Pb in Mokola had a CF of 0.49. The highest total colony count was found in Akomoje ( $10.0 \times 10^5$ ), with the total coliform count being highest in Saje ( $0.8 \times 10^5$ ), Staphylococcus aureus count in Saje ( $1.2 \times 10^5$ ), Bacillus spp. counts in both Saje and Mokola ( $0.4 \times 10^5$ ), Pseudomonas spp. count in Akomoje ( $0.2 \times 10^5$ ), and Micrococcus spp. count across all sites ( $0.1 \times 10^5$ ). The Serratia marcescens count in Akomoje was  $0.3 \times 10^5$ . Furthermore, the total fungi count was highest in Saje ( $0.5 \times 10^5$ ), with Saccharomyces spp. count in Akomoje at ( $0.8 \times 10^5$ ), Aspergillus fumigatus count in Obantoko ( $0.2 \times 10^5$ ), and Aspergillus viridis count in Obantoko ( $0.3 \times 10^5$ ). In conclusion, this study indicates significant potential environmental and public health risks, leading to a decline in environmental quality.*

**Keywords:** Heavy Metals; Microbial Load; Top Soil; Dumpsites; Public Health.



## Evaluating the Compressive and Tensile Strength of Hybrid Foam Concrete Formulated with A Palm Kernel Oil-Based Surfactant

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### Abstract

*The growing demand for sustainable and lightweight construction materials has driven interest in hybrid foam concrete, which enhances the load-bearing capacity of traditional foam concrete by incorporating aggregates like stone dust. This study provides a focused evaluation of the mechanical properties of a hybrid foam concrete mix formulated with a sustainable, palm kernel oil-based surfactant (PKOS). The experimental program involved producing concrete specimens using a 1:2 mix ratio (cement to aggregates) and a high water-to-cement ratio of 0.75. The mix design incorporated a 0.5% concentration of PKOS by weight of cement alongside Sodium Lauryl Sulfate (SLS) as the primary foaming agent. Specimens were cast as 150 mm cubes for compressive strength and density analysis, and 110 mm diameter cylinders for splitting tensile strength analysis, followed by a standard water curing regimen for 7, 14, and 28 days. The results demonstrate that the PKOS-stabilized system is effective in producing lightweight concrete, with densities decreasing to as low as 1828 kg/m<sup>3</sup> for cubes at 28 days. However, the mechanical performance was modest and exhibited an anomalous trend. Compressive strength peaked at 14 days, reaching approximately 2.2 N/mm<sup>2</sup>, before declining to 1.5 N/mm<sup>2</sup> at 28 days. A similar trend was observed for splitting tensile strength. This strength regression suggests potential long-term microstructural instability in the high water-content mix. The study concludes that while PKOS is a viable bio-based surfactant for achieving low density, its application requires careful optimization of the overall mix design, particularly the water-to-cement ratio, to ensure durable, long-term mechanical performance. The findings classify the material as suitable for non-load-bearing applications.*

**Keywords:** Palm kernel oil, Compressive Strength, Splitting Tensile Strength, Lightweight Concrete, Sustainable Materials, Sodium Lauryl Sulfate (SLS).





## Comparative Study of Traditional On-Premise IT And Cloud Computing: A Documentary Research Perspective

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### Abstract

*This study conducts a comparative analysis of Traditional On-Premise IT and cloud computing by synthesizing insights from twenty peer-reviewed journal articles, industry blogs, and technical reports. The research employs qualitative content analysis to systematically extract, code, and tabulate data from diverse scholarly and professional sources. The two models were compared using performance indicators such as, cost, scalability, speed of deployment, efficiency in operation, security, and integration capabilities. Results show that traditional cloud computing provides established infrastructure, stability, and performance predictability for large-scale enterprise systems; while cloud computing has better scalability, deployment speed, developer experience and entry costs for small-medium projects. The research also points out the latency tradeoffs, customization flexibility and other vendor lock-in risks. This work synthesizes evidenced across multiple studies to yield a consolidated knowledge base that inform decision-making for organizations moving to cloud-based solutions. Additionally, the research illustrates which models become preferable due to context-specific needs (e. g., workload type, compliance requirements and long-term cost-related aspects) that significantly impact the selection between both à la carte types. The findings help advance the IS literature by providing a holistic view and a structured assessment framework to orient practitioners, researchers and policy makers toward optimal cloud deployment alternatives.*

**Keywords:** Cloud computing, Comparative analysis, Content analysis, Qualitative research, Web application.



## Urban Liveability and Social Exclusion: A Bibliometric Review

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### Abstract

*A thorough bibliometric study of the literature on social exclusion and urban liveability is carried out in this review. The study attempts to pinpoint important topics, trends, and areas in need of further research in the discipline by methodically going over many academic publications. The analysis highlights the need for a comprehensive understanding of urban dynamics by revealing a growing body of research on the complex relationship between social exclusion and urban liveability. The review offers insights into the changing conversation surrounding urban surroundings by identifying popular subjects and research approaches. The analysis recognizes the advancements in the discipline, but it also highlights other areas that are still understudied and need more research. These deficiencies include the requirement for interdisciplinary approaches, a wider range of viewpoints, and thorough research to address the complexity present in urban environments. In addition to being a useful tool for scholars attempting to navigate the body of literature, this bibliometric approach has applications for communities, urban planners, and legislators. The results highlight the significance of inclusive urban development and add to a more nuanced knowledge of the potential problems related to social exclusion and urban liveability. This review concludes by synthesizing the body of knowledge, pointing out areas of research interest, and recommending further research into the field of urban studies. Combining the best available data lays the groundwork for further studies that can support evidence-based policymaking and create liveable, socially inclusive cities for a variety of demographics.*

**Keywords:** Urban liveability, Social exclusion, Bibliometric, Environments, Policymaking.



## Isolation and Characterization of Lactic Acid Bacteria from Cassava Fermentation to Produce Gari

By

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### Abstract

*Isolation and characterization of microorganisms from fermented cassava are essential steps in understanding the microbial diversity and dynamics involved in gari production. By studying these microorganisms, researchers can identify potential starter cultures for improved gari production, enhance food safety, and optimize fermentation conditions to achieve consistent product quality. This aims of this project is to isolate, characterize and identify the microorganisms in fermented cassava used for gari production. The study employed traditional microbiological techniques (conventional method). The specific objectives include: i. to isolate the lactic acid bacteria species, present during the fermentation of cassava to produce gari, ii. to characterize the lactic acid bacteria isolates involved in the fermentation. The isolates were suspected to be Weissella sp., Enterococcus sp., and Pediococcus sp. The findings from this research contribute to a deeper understanding of the involvement of LAB in the fermentation process of cassava to produced fermented gari. More research is required to confirm the potential of fermented gari in the treatment of diarrhea.*



## Antidiarrhea Effect of Lactic Acid Bacteria from Cassava Fermentation to Produce Gari in Albino Rats

By

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### Abstract

*Understanding the involvement of lactic acid bacteria (LAB) in the fermentation of cassava to produce gari is essential for ensuring product potential to alleviate the occurrence of diarrhea. By studying the presence of these microorganisms, researchers can confirm the potential of gari for improved diarrhea treatment as an alternative to treatment of diarrhea using antibiotics. The objectives of this project is to isolate, identify and examine the antidiarrhea effect of gari in albino rats. The study employed molecular method to identify the lactic acid bacteria and examine their antidiarrhea effect in albino rats with E. coli induced diarrhea. The isolates were identified to be Weissella cibaria., Enterococcus durans., and Pediococcus pentosaceus. The findings from this research contribute to a deeper understanding of the involvement of LAB in the fermentation process of cassava to produced fermented gari and potential of gari to ameliorate diarrhea. More research is required to confirm the potential of other fermented foods in the treatment of diarrhea.*



## Enhancing the Engineering Properties of Lateritic Soil with Recycled Asphalt for Road Pavement Development

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### Abstract

*This paper investigates the stabilization of lateritic soils using recycled asphalt materials (RAM) for road pavement applications. Lateritic soils in tropical regions are widely used but often suffer from low strength and high plasticity, which contribute to early pavement failure. Conventional stabilizers such as cement and lime are effective but costly and environmentally unsustainable. To address this, lateritic soil samples were blended with RAM at 0%, 5%, 10%, 15% and 20% and tested for atterberg limits, compaction, California Bearing Ratio. The results demonstrate that RAM significantly enhances soil strength, with optimum performance at 15% RAM yielding a soaked CBR of 24%. Both values exceed the minimum requirements for subgrade and base course applications. In addition to technical improvements, the use of RAM reduced construction costs and lower the green house emissions. These finding suggest RAM is a sustainable and cost - effective alternative for pavement construction.*

**Keywords:** Lateritic Soil, Recycled Asphalt Materials, Soil Stabilization, California Bearing Ratio, compaction test, triaxial test, Sustainable Pavement Engineering.



## The Effects of Climate Change on Post-Harvest Losses of Fruits and Vegetables in Nigeria: A Review

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### Abstract

*This paper reviewed the effects of climate change on post-harvest loss of fruits and vegetables in Nigeria. The specific objectives were to identify the causes of climate change; identify the causes of post-harvest losses; identify the effects of climate change on post-harvest losses; and proffer solutions to the climate changes on the post-harvest losses in Nigeria. The result reveals that the causes of climate change are volcanic eruptions, ocean currents, meteorite impacts, solar variations carbon-di-oxide, nitrous oxide, chlorofluorocarbon, burning of fossil fuels, gas flaring, and deforestation. Also, findings revealed that the causes of post-harvest losses include biological, chemical, mechanical, physiological, microbiological, improper post-harvest handling practices, inadequate facilities and necessary infrastructure, improper temperature management, inadequate of management knowledge/skill, inappropriate packaging, inadequate transportation facilities, inadequate marketing systems and government regulations and legislations. The effects of climate change on post-harvest losses includes global warming, higher temperatures, higher humidity, unseasonal rains, to address these challenges of climate changes on post-harvest losses, several mitigation strategies, such as strengthening research and development institutions, improve access to information and capacity building, improve on farm post-harvest practices and governments' policies improvement for postharvest loss reduction. The review concluded that climate change seriously affects all the stages of food value chains of fruits and vegetables. The paper recommended that reducing food loss through improving storage technologies/ facilities can increase climate resilience and can reduce the vulnerability of fruits and vegetables to heat or pest infestation.*

**Keywords:** Climate-Change, postharvest losses, and Nigeria.





## The Fundamentals of Capturing Light from Individual Photosynthetic Complexes

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### Abstract

*The design of bio-inspired technologies and biological research relies on understanding how photosynthetic organisms absorb and transform light energy at the molecular level. Recent developments in spectroscopy, microscopy, and single-molecule techniques have enabled studies of how distinct photosynthetic complexes collect photons, transmit excitonic energy, and guide it towards reaction centres. This study examines the fundamentals of how these complexes catch light, the structural and functional arrangement of pigment-protein assemblies, the significance of quantum coherence, and the new uses of this information in nanotechnology and artificial photosynthesis. Photosynthetic systems harness sunlight to power most life on Earth. In the initial steps of photosynthetic light harvesting, absorbed energy is converted to chemical energy with near-unity quantum efficiency. Three key principles of this flow and of photosynthetic light harvesting are discussed: thermal fluctuations of the protein structure, intrinsic conformational switches with defined functional consequences, and environmentally triggered conformational switches. Understanding how molecular machinery and dynamics are designed to balance these costs may provide a blueprint for improved artificial light-harvesting devices. A multi-disciplinary approach combining knowledge of biology could lead to low-cost and more effective solar energy conversion.*

**Keywords:** Light, photosynthetic, thermal fluctuation, bio-inspired, earth.



## Biophysics of Solar Energy Materials Influenced by Biology

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### Abstract

*The study discusses the role of biology in the design and understanding of solar energy materials, focusing on the physical principles underlying biological light capture. It also discusses the potential of using biological solar energy as a substitute for fossil fuels in energy generation. The paper also explores the use of artificial photosynthesis as a means of biological solar energy, which directly converts sunlight into fuel and electricity. This energy source can be harnessed by designing a prototype sun or creating artificial plants that efficiently capture sunlight for the photosynthetic process. This energy source has the potential to supply the world's energy needs and combat climate change if harnessed properly. The study highlights the importance of balancing the light-dependent reaction and light-independent reaction in the development of sustainable energy sources.*

**Keywords:** *Biological Solar Energy; Artificial Photosynthesis, Biomass, Light–Dependent Reaction; Light-Independent Reaction; Sustainable Energy.*



## Evaluation of Antioxidant Activity of Methanolic Extracts of Leaf of *Cnidoscolus Aconitifolius*

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### Abstract

*This study investigated the antioxidant potential of methanolic leaf extracts of Cnidoscolus aconitifolius using standard in vitro assays, including DPPH, nitric oxide, lipid peroxidation, and superoxide anion scavenging tests. The extracts exhibited notable radical scavenging activity, with IC<sub>50</sub> values for the leaf extract recorded at 30.94 µg/ml (DPPH), 24.70 µg/ml (lipid peroxidation), 40.03 µg/ml (nitric oxide), and 26.38 µg/ml (superoxide anion), closely matching or complementing the activity of ascorbic acid. While the reference antioxidant (ascorbic acid) consistently showed higher efficacy, the extracts demonstrated the inhibition of oxidative radicals. These findings suggest that Cnidoscolus aconitifolius possesses substantial antioxidant activity attributable to its bioactive compounds, supporting its traditional use and potential application in the development of natural therapeutics against oxidative stress-related disorders.*

**Keywords:** *Cnidoscolus aconitifolius, antioxidant activity, DPPH assay, lipid peroxidation.*



## A Study on The Viability of Utilizing Palm Kernel Shells as A Replacement for Conventional Aggregates in Concrete Production

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### Abstract

*The rising cost of conventional construction materials such as cement and granite in developing countries necessitates the exploration of sustainable and cost-effective alternatives. Palm kernel shells (PKS), an abundant agricultural waste product in Nigeria, offer a promising solution. This study investigates the feasibility of using PKS as a complete replacement for coarse aggregates in the production of concrete interlocking paving blocks. The research methodology involved characterizing the physical properties of PKS, producing PKS concrete with a 1:1:2 mix ratio and a 0.5 water-cement ratio, and testing the compressive strength of the concrete at intervals of 7, 14, and 28 days. Subsequently, interlocking paving stones were fabricated from the PKS concrete and installed on a test pavement at the University of Ilorin. The results indicated that while PKS has a lower specific gravity and higher water absorption capacity compared to traditional aggregates, the produced concrete met the required strength for lightweight applications. Adjustments in the mix design successfully compensated for the material properties of PKS, resulting in interlocking stones with adequate durability and performance. The study concludes that PKS is a viable substitute for conventional aggregates in concrete, offering significant cost savings and environmental benefits by repurposing agricultural waste. This approach provides a sustainable and economically feasible alternative for pavement construction in regions with abundant PKS availability.*

**Keywords:** Palm kernel shells, concrete interlocking paving blocks, sustainable construction, lightweight concrete, cost-effective materials.



## Performance Evaluation of Concrete Incorporating Crushed Waste Glass as Partial Coarse Aggregate Replacement

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### Abstract

*The rising cost of conventional aggregates and the environmental challenges associated with disposing of non-biodegradable waste highlight the need for sustainable alternatives in construction. This study explores the potential of incorporating crushed waste glass as a partial substitute for natural coarse aggregates in concrete. The primary aim was to evaluate the impact of this substitution on the workability and compressive strength of concrete. Experimental tests were conducted by replacing coarse aggregates with crushed glass at proportions of 0% (control), 5%, 10%, 15%, and 20% by weight. Concrete specimens were prepared using a mix ratio of 1:2:4 and a water–cement ratio of 0.5. Standard slump tests and compressive strength tests were carried out after 7, 14, and 28 days of curing. Findings revealed that increasing the proportion of crushed glass enhanced the workability of the mixes but resulted in a reduction in density and compressive strength. At 28 days, the control mix attained a compressive strength of 17.19 N/mm<sup>2</sup>, while the 5% replacement mix recorded a comparable value of 16.11 N/mm<sup>2</sup>. Higher replacement levels showed more pronounced strength losses. The study concludes that a 5% substitution of coarse aggregates with crushed waste glass provides an optimal balance, yielding lightweight concrete with minimal strength reduction, thereby offering a sustainable solution for both construction and waste management.*

**Keywords:** Crushed waste glass, Coarse aggregate replacement, Workability, Compressive strength, Lightweight concrete, Sustainable construction, Waste management.



## Study of Neem Oil Extract as an Insecticide Against Beans Weevil (*Acanthoscelides Obtectus*) in Storage Beans

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### Abstract

*The infestation of stored beans by bean weevil (*Acanthoscelides Obtectus*) poses significant threats to food security particularly in developing countries. Synthetic insecticides are widely used but raise concerns over environmental pollution, human health hazards, and insect resistance. This study aims to evaluate the efficacy of neem (*Azadirachta indica*) oil extract as a biopesticides against bean weevil (*A. Obtectus*) in stored beans. A laboratory experiment was conducted, in which 200gm of infested bean samples were treated with neem oil at Different concentrations 1ml, 2ml and 3ml and 200gm infested bean samples were set as control, the experiment was laid out in a completely randomised design with four replications to assessed adult mortality and seed damage. The results revealed that pure neem oils were applied to infested bean samples at various concentration adequately treat the infested beans grains compere with control. These findings provide evidence of the potential of neem oil as a safer and eco-friendly alternative to synthetic insecticides for protecting stored beans.*

**Keywords:** neem oil, infestation beans, beans weevil.



## Design and Fabrication of a Water Dispenser for Household Use

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### Abstract

*Voluminous consumption of bottled water daily spurs the use of water dispensers and coolers at homes and in offices. In the Cities vending machines are growing in numbers, mostly especially in conspicuous places such as: parks and schools – dispensing cold water and even hot coffee or chocolates. Health compliance and other issues such as maintenance of the units become a concern that the city government has to address. Access to clean water is a fundamental human right, and a well-designed water dispenser contributes to fulfilling this right. By fabricating an efficient and cost-effective water dispenser, this study aims to address water accessibility challenges in various societal settings, promoting health and well-being. The fabrication process involves the integration of innovative technologies to enhance the functionality of the water dispenser. This study contributes to the advancement of water dispensing systems, potentially influencing future designs and implementations.*





## Engineering and Technology Education for Sustainable Development in Nigeria

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### Abstract

*Engineering and Technological Education remains vital for robust economic growth and sustainable development. It fosters advanced knowledge, habit, and skills, transferable from one generation to the next through training, research, and teaching. This paper examines the economic challenges facing Nigeria, such as lethargic global economic integration, declining technological capacity, and inadequate skill acquisition - resulting from poor engineering and technology education. Relying on academic literature and policy analysis, the study emphasized the need for quality education, continuous workforce training, and capacity building in response to disruptive technologies. Integrating skilled labor into local and global industries is crucial for national development and economic resilience.*

**Keywords:** Technology education, Capacity building, Sustainable development, National economy.



## Enhancing Examination Integrity and Grading in Nigeria through AI-Powered Assessment Tools

By

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### Abstract

*In Nigeria, challenges such as examination malpractice, inconsistent grading, and human biases continue to undermine the integrity and reliability of educational assessments. This paper explores how Artificial Intelligence (AI)-powered assessment tools can be leveraged to enhance examination integrity and improve grading processes within the Nigerian educational system. By utilizing technologies such as automated proctoring, machine learning-based grading systems, and intelligent plagiarism detection, AI offers scalable and transparent solutions to combat examination fraud, ensure fair evaluation, and provide timely feedback. This study highlights case examples and potential implementation models suitable for Nigerian institutions, while also addressing concerns around data privacy, infrastructural readiness, and stakeholder acceptance. Ultimately, the paper argues that integrating AI into assessment practices can significantly strengthen trust in Nigeria's educational outcomes, foster academic excellence, and align the system with global standards.*

**Keywords:** Artificial Intelligence (AI), Educational Assessment, Examination Integrity, Grading Systems, Automated Proctoring, Machine Learning



## Antibacterial Activities and HPLC Analysis of the Leaf Extract of *Martynia Annu* Against Multidrug Resistance Bacteria

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### Abstract

*This study evaluates the antibacterial activities and HPLC analysis of the leaf extract of Martynia annua against multidrug resistance strain of Acinetobacter baumannii and Klebsiella pneumoniae. Initial antibiotic susceptibility testing revealed that A. baumannii was resistant to all tested antibiotics, while K. pneumoniae displayed susceptibility to ofloxacin and levofloxacin, with inhibition zones ranging from 26-29 mm. Antibacterial testing of M. annua leaves extracts demonstrated variable efficacy. The methanolic extract exhibited the most substantial antibacterial activity, with inhibition zones for K. pneumoniae ranging from 15-20 mm at 4 and 8 µg/ml, while A. baumannii showed inhibition zones ranging from 10-20 mm. The ethyl acetate extract displayed inhibition zones of 14-22 mm for A. baumannii at all tested concentrations, though K. pneumoniae did not show clear zones. The n-hexane extract demonstrated activity against A. baumannii at higher concentrations (32 and 64 µg/ml), with zones of 16-17 mm, but was ineffective at lower concentrations. Minimum Inhibitory Concentration (MIC) tests revealed that both bacteria showed no growth at 32 and 64 µg/ml for the ethanolic and n-hexane extracts. Minimum bactericidal concentration (MBC) results indicated bactericidal effects at 64 µg/ml for A. baumannii in ethanol and n-hexane, while K. pneumoniae displayed growth under similar conditions. Qualitative phytochemical analysis of the extracts revealed the presence of tannins, flavonoids, alkaloids, phlobatannins, terpenoids, steroids, and coumarins. High- Performance Liquid Chromatography (HPLC) analysis confirmed the presence of bioactive compounds such as catechin, quercetin, rutin, and luteolin. The antioxidant activity, assessed using DPPH assay, demonstrated notable free radical scavenging capacity, with ethanolic extract showing values*



*ranging from 87.0813 to 90.4306% at varying concentrations (100-500 mg/ml), and ascorbic acid (reference standard) ranging from 86.4692 to 94.3155%. These findings highlight the potential of *M annua* leaves extract as a source of bioactive compounds with antibacterial and antioxidant properties.*

**Keywords:** *Martynia annua*, multi-drug resistance, antibacterial activities, antioxidant properties



## Evaluation of the Phytochemical Constituents and Synergistic Antibacterial Activities of *Martynia Annu* Leaf Extract Against Multi-Drug Resistant Pathogens

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### Abstract

*This study evaluates the antibacterial activities and HPLC analysis of the leaf extract of Martynia annua against multidrug resistance bacteria, including Pseudomonas aeruginosa and Enterobacter cloacae. Initial antibiotic susceptibility testing revealed that P. aeruginosa was resistant to all tested antibiotics, while E. cloacae displayed susceptibility to Ampicillin plus cloxacillin with inhibition zones 10mm. Antibacterial testing of M. annua leaves extracts demonstrated variable efficacy. The combination of methanolic, ethyl-acetate, and N-hexane extract revealed that Pseudomonas aeruginosa showed zone of inhibition at 100µg and 200µg were 11mm-12mm, while there was no zone of inhibition recorded at 500µg. The zone of inhibition of Enterobacter cloacae at 100µg, 200µg and 500µg was 16mm, 15mm, and 14mm respectively. Minimum Inhibitory Concentration (MIC) tests revealed that both bacteria showed no growth at 32 and 64 µg/ml for the ethanolic and n-hexane extracts. Minimum bactericidal concentration (MBC) results indicated bactericidal effects at 64 µg/ml for P. aeruginosa in ethanol and n-hexane, while E. cloacae displayed growth under similar conditions. Qualitative phytochemical analysis of the extracts revealed the presence of tannins, flavonoids, alkaloids, phlobatannins, terpenoids, steroids, and coumarins. High- Performance Liquid Chromatography (HPLC) analysis confirmed the presence of bioactive compounds such as catechin, quercetin, rutin, and luteolin. The antioxidant activity, assessed using DPPH assay, demonstrated notable free radical scavenging capacity, with ethanolic extract showing values ranging from 87.0813 to 90.4306% at varying concentrations (100-500µg/ml), and ascorbic acid ranging from 86.4692 to*



94.3155%. These findings highlight the potential of *M annua* leaves extract as a source of bioactive compounds with antibacterial and antioxidant properties.

**Keywords** – *Martynia annua*, multi-drug resistance, *Pseudomonas aeruginosa*, *Enterobacter cloacae*



## English for Specific Purposes in the Age of AI: A Multiliteracies Perspective

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### Abstract

*This paper explores the crucial integration of Artificial Intelligence (AI) tools into English for Specific Purposes (ESP) instruction through the lens of the Multiliteracies Perspective. In today's emerging digital ecosystem, traditional ESP approaches often fall short in equipping learners for domain-specific communication in technologically advanced fields. We employed a qualitative descriptive approach to investigate how AI-powered applications like ChatGPT, Grammarly, and ELSA Speak can significantly enhance language learning for contexts such as marine, technical and other professions. Our analysis of existing ESP curricula and instructional resources reveals a limited integration of digital tools and multimodal learning. However, our findings highlight the immense potential of AI tools to foster learner autonomy, critical thinking and multimodal fluency. We also identify key challenges, including infrastructure limitations, the need for enhanced teacher preparedness, and effective content localization. We argue for a fundamental redesign of ESP curricula to fully embrace multiliteracies and AI integration. This necessitates strategic institutional investment in comprehensive teacher training, supportive policy reforms, and equitable access strategies. In conclusion, we position AI-enhanced, multiliteracies-informed ESP as indispensable for preparing learners for contemporary workplace communication and contributing to sustainable development.*

**Keywords :** ESP, Artificial Intelligence, Multiliteracies.





## Direct Bioautography for the Screening of the Antibacterial Properties of the Leaf Extract of *Macaranga Barteri* Against Multi-Drug Resistance Strain of Clinical Origin

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### Abstract

*The escalating threat of multidrug-resistant (MDR) bacteria underscores the need for new antibacterial agents from natural sources. This study investigated the in vitro antibacterial activity of stem extracts from Macaranga barteri, a plant traditionally used to treat infectious diseases. The extracts were prepared using maceration with n-hexane, chloroform, ethyl acetate, methanol, and acetone. Their efficacy was evaluated against clinical isolates of MDR strains of Staphylococcus aureus and Enterobacter species. Direct bioautography was used to screen the antibacterial properties. All five extracts showed varying degrees of inhibitory activity, with the ethyl acetate extract demonstrating the largest zones of inhibition (1.0–1.6 mm) against the tested bacteria. In contrast, the methanol extract showed inhibition zones ranging from 0.1–1.1 mm, n-hexane from 0.2–0.8 mm, acetone from 0.5–1.1 mm, and chloroform from 0.3–0.9 mm. Despite these observable inhibitory effects in the bioautography assay, the stem extracts of M. barteri did not demonstrate significant antibacterial activity against the tested MDR strains, suggesting that the bacteria possess resistance mechanisms capable of overcoming the extracts' potential inhibitory effects. Phytochemical screening revealed the presence of tannins, saponins, flavonoids, alkaloids, glycosides, and terpenoids, which are known to have biological activity. In conclusion, although the M. barteri extracts contain bioactive compounds, they do not appear to be effective against the specific MDR strains of S. aureus and Enterobacter species used in this study. This highlights the complexity of developing plant-based antimicrobials and the potential for bacterial resistance to natural compounds.*

**Keywords:** Multidrug-resistant bacteria, *Macaranga barteri*, Antibacterial activity, Direct bioautography.



## Evaluation of the Phytochemical Constituents and Synergistic Antibacterial Activities of *Martynia Annu* Leaf Extract Against Multi-Drug Resistant Pathogens

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### Abstract

*This study evaluates the antibacterial activities and HPLC analysis of the leaf extract of Martynia annua against multidrug resistance bacteria, including Pseudomonas aeruginosa and Enterobacter cloacae. Initial antibiotic susceptibility testing revealed that P. aeruginosa was resistant to all tested antibiotics, while E. cloacae displayed susceptibility to Ampicillin plus cloxacillin with inhibition zones 10mm. Antibacterial testing of M. annua leaves extracts demonstrated variable efficacy. The combination of methanolic, ethyl-acetate, and N-hexane extract revealed that Pseudomonas aeruginosa showed zone of inhibition at 100µg and 200µg were 11mm-12mm, while there was no zone of inhibition recorded at 500µg. The zone of inhibition of Enterobacter cloacae at 100µg, 200µg and 500µg was 16mm, 15mm, and 14mm respectively. Minimum Inhibitory Concentration (MIC) tests revealed that both bacteria showed no growth at 32 and 64 µg/ml for the ethanolic and n-hexane extracts. Minimum bactericidal concentration (MBC) results indicated bactericidal effects at 64 µg/ml for P. aeruginosa in ethanol and n-hexane, while E. cloacae displayed growth under similar conditions. Qualitative phytochemical analysis of the extracts revealed the presence of tannins, flavonoids, alkaloids, phlobatannins, terpenoids, steroids, and coumarins. High- Performance Liquid Chromatography (HPLC) analysis confirmed the presence of bioactive compounds such as catechin, quercetin, rutin, and luteolin. The antioxidant activity, assessed using DPPH assay, demonstrated notable free radical scavenging capacity, with ethanolic extract showing values ranging from 87.0813 to 90.4306% at varying concentrations (100-500µg/ml), and ascorbic acid ranging from 86.4692 to*



94.3155%. These findings highlight the potential of *M annua* leaves extract as a source of bioactive compounds with antibacterial and antioxidant properties.

**Keywords** – *Martynia annua*, multi-drug resistance, *Pseudomonas aeruginosa*, *Enterobacter cloacae*



## Phytochemical Analysis and Antibacterial Synergy of *Martynia Annua* Leaf and Root Extracts Against Multi Drug Resistance Bacteria Evaluation of the Phytochemical Constituents and Synergistic Antibacterial Activities of *Martynia Annua* Leaf Extract Against Multi-Drug Resistant Pathogens

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### Abstract

*The escalating global health crisis posed by multidrug-resistant (MDR) pathogens necessitates the urgent discovery of new therapeutic agents. This study investigated the antibacterial and antioxidant properties of leaf extracts from Martynia annua against two significant MDR pathogens, Klebsiella pneumoniae and Acinetobacter baumannii. Antibiotic susceptibility testing confirmed that A. baumannii was resistant to all tested antibiotics, while K. pneumoniae was susceptible to Levofloxacin and Ofloxacin, showing inhibition zones of 26-29 mm. The study employed methanolic, ethyl acetate, and n-hexane extracts of M. annua. Antibacterial efficacy was evaluated using Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) assays. The ethanolic and n-hexane extracts demonstrated promising inhibitory effects, with no bacterial growth observed at 32 and 64 µg/ml for both species. The MBC results were particularly noteworthy, as the extracts exhibited a bactericidal effect on A. baumannii at 64 µg/ml, whereas K. pneumoniae remained viable under the same conditions. Phytochemical analysis revealed the presence of several bioactive compounds, including tannins, flavonoids, alkaloids, and terpenoids. High-Performance Liquid Chromatography (HPLC) further confirmed the presence of specific compounds like catechin, quercetin, rutin, and luteolin. Furthermore, the ethanolic extract showed potent antioxidant activity, with a DPPH free radical scavenging capacity ranging from 87.08% to 90.43% at 100-500 ng/ml, comparable to the ascorbic acid standard (86.46% to 94.31%). These findings underscore the significant potential of M. annua leaves as a source of novel antibacterial and antioxidant agents. This study contributes to the growing body of evidence supporting the use of plant-based therapeutics to combat MDR pathogens and provides valuable insights into the pharmacological potential of Martynia annua.*

**Keywords:** Multidrug-resistant bacteria, Martynia annua, Antibacterial activity, Phytochemical analysis.



## AI Model for Inflation Prediction: A Macroeconomic Indicator-Based Approach in Nigeria

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### Abstract

*This paper presents the design and implementation of an Artificial Intelligence (AI)-based model for forecasting inflation in the Nigerian economy. Monthly macroeconomic data from 2000 to 2024 were sourced from the Central Bank of Nigeria, National Bureau of Statistics, International Monetary Fund, and World Bank. Data pre-processing involved handling missing values, normalization, outlier detection, and feature engineering. Four models, Autoregressive Integrated Moving Average (ARIMA), Random Forest, Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM) neural networks, were developed and evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Results show that AI-based models consistently outperformed ARIMA, with the LSTM model achieving the best performance (RMSE = 0.87, MAE = 0.75, MAPE = 4.7%). The findings demonstrate the superiority of AI approaches in capturing nonlinear and temporal dependencies in Nigeria's inflation dynamics. This research contributes to the development of evidence-based forecasting tools that can enhance monetary policy, economic planning, and financial decision-making in Nigeria. Future work should focus on integrating high-frequency indicators, incorporating explainable AI techniques, and deploying real-time dashboards for policymakers.*

**Keywords:** Inflation Forecasting, Artificial Intelligence, Machine Learning, Nigerian Economy, ARIMA, Economic Policy



## A Review on Food Innovations with Microbes for Ensuring Food Security

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### Abstract

*Food security occurs when people have the social and economic means to physical access to safe and nutritious food that meets their dietary necessities and preferences for the maintenance of an active and healthy life, at all times. Food spoilage leads to significant waste, with approximately 14% of the world's food lost before reaching the market and this loss exacerbates food insecurity. Perishable foods, including fresh produce, dairy, and meats, have inherently short shelf lives, posing challenges for storage and distribution. The rapid deterioration of these items often results in increased food waste and economic losses. Efforts to extend the shelf life of perishable Foods are essential to enhance food security and reduce waste. Recent innovations leveraging microbes have significantly enhanced food security by improving food preservation, nutritional value, and safety. The use of bacteriocins as natural preservatives, functional foods enriched with probiotics, and microbial enzymes in food processing has contributed to healthier and more sustainable food systems. Hurdle technology, which combines multiple preservation techniques, has effectively extended shelf life, while microbial fortification has addressed micronutrient deficiencies. These microbial-based innovations offer environmentally friendly and efficient solutions to combat food spoilage, malnutrition, and foodborne diseases, making them vital tools in ensuring global food security. To maximize the potential of microbial food innovations, governments and research institutions should invest in advanced microbial biotechnology research that promote safe use of microbial-derived food components. Food industries should integrate microbial fortification, hurdle technology, and natural preservatives into production to enhance food quality while reducing waste.*

**Keywords:** Food security, innovations, biotechnology, microbes.



## Re-Evaluating Weeds: Ecological and Economic Relevance

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### Abstract

*Weeds are referred to as unwanted and undesired plants growing competitively. Weeds are known to reduce agricultural productivity and cause significant losses. In every ecological habitat, there are diverse weed species and ecotypes adapted to different of environmental conditions and are ready to proliferate when they receive proper signal. A foot long size of soil, 6 inches deep, can hold from 98 to 3,068 of viable weed seeds. The period a seed is capable of producing a seedling depends on its viability, varies with weed species. Dormant seeds of weeds such as *Striga hermonthica* and *Datura stramonium* have over a 90 percent germination rate even after some years in the soil. Despite the ravaging damages weeds can cause to crops, many weed species play significant ecological and economic functions. This paper highlights established and often neglected benefits of weeds, which range from forage, traditional/folklore medicine, soil fertility and erosion control, and even for bioremediation and eutrophication purposes. *Talinum triangulare* is a weed which has culinary value as a vegetable and also a potherb. *Chrysanthemum cinerariifolium*, a weed produces pyrethrum, which is a natural insecticide that has low toxicity and applied widely in organic farming as a source of pest control. Pyrethrum is potent against some insects and has low toxicity to humans and animals thus can serve as a better eco-friendly option over synthetic chemicals. *Cynodon dactylon*, is a weed that serves as ground cover, with*





*horticultural visual appeal and effective against erosion due to the soil-binding strength. This re-evaluation study of the role of weeds in ecosystems will create a more unbiased approach to weed management.*

**Keywords:** *Weeds, ecotypes, ecological, evaluation, eutrophication*



## Utilizing AI-Powered Services for Proactive Cybersecurity

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### Abstract

*The rapid evolution of cyber threats has rendered traditional reactive cybersecurity measures insufficient in combating sophisticated attacks. This necessitates a shift towards proactive defense mechanisms, leveraging artificial intelligence (AI) and machine learning (ML) to predict, detect, and neutralize threats before they escalate. AI-powered cybersecurity solutions, including predictive analytics, automated threat intelligence, and anomaly detection, offer real-time risk identification and faster, more accurate responses to security incidents. These technologies enhance digital defenses by continuously learning from vast datasets, identifying emerging attack patterns, and adapting security protocols accordingly. This paper explores the transformative role of AI in cybersecurity, emphasizing the advantages of shifting from a reactive to a proactive security approach. While AI-driven solutions improve threat detection, response times, and automation, they also introduce challenges such as false positives, adversarial AI attacks, and ethical concerns regarding data privacy and decision-making transparency. Addressing these challenges is crucial for maximizing the potential of AI in cybersecurity. By adopting AI-powered security frameworks, organizations can anticipate, prevent, and mitigate cyber threats more effectively, thereby ensuring a safer and more resilient digital environment. This research highlights the benefits and limitations of AI in cybersecurity, offering insights into how businesses and institutions can integrate AI-driven solutions to strengthen their security posture and minimize risks.*

**Keywords:** Artificial intelligence (AI), Proactive, Cyber threat, Machine Learning (ML), Detection



## Advancing Sustainable AI through Efficient Hyperparameter Tuning Techniques

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### Abstract

*Effective parameter tuning is fundamental to optimizing machine learning (ML) and artificial intelligence (AI) models, as it directly impacts their adaptability, generalization capability, and predictive accuracy. Hyperparameters—such as learning rate, tree depth, and the number of estimators—must be carefully selected to prevent underfitting or overfitting, minimize computational overhead, and ensure model robustness. This paper presents a critical review of parameter tuning methodologies, encompassing traditional techniques like manual and grid search, probabilistic strategies such as Bayesian optimization, and advanced meta-heuristic approaches inspired by evolutionary processes. The discussion extends to the strengths, limitations, and scalability of these methods, emphasizing their relevance to real-world, data-driven systems. Furthermore, the paper explores the role of automated parameter tuning in fostering sustainable development by enabling efficient resource utilization, reducing computational waste, and improving decision-making in sectors such as healthcare, finance, and smart education. By providing a comprehensive analysis of tuning strategies, this study aims to guide researchers and practitioners in developing optimized AI models that support sustainable and responsible innovation.*

**Keywords:** Hyperparameter optimization, Bayesian optimization, sustainable AI, automated parameter tuning, machine learning efficiency.



## Evaluation of Electroactivity of a Local *Fungal* Isolate from Pond Water

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### Abstract

*Power supply continues to be a significant challenge in developing nations including Nigeria. This study aimed at assessing the potential of a fungus for generating bioelectricity using microbial fuel cells. The fungus was isolated from pond water using the pour plate method and identified tentatively based on its morphological traits, revealing it to be Aspergillus sp. This isolated fungus was then tested alone for bioelectricity production in microbial fuel cells, with cassava wastewater considered as a potential substrate. The microbial fuel cells were set up in an H-shaped configuration. Open Circuit Voltage readings were taken every six hours over the course of seven days, and daily averages were calculated. On the first day, the organism produced 804.5 mV of Open Circuit Voltage, which increased to a peak of 1266.6 mV by day six. A decline in the open circuit voltage produced was recorded on day seven. The findings of this study indicate that, when effectively utilized, microbial fuel cell (MFC) technology has the potential to provide an alternative source of renewable energy and fungi have the potential for bioelectricity production in microbial fuel cells.*

**Keywords:** *Microbial fuel cells, electroactivity, cassava wastewater, fungus, open circuit voltage.*



## Air Pollution and Its Effects on the Human Immune System

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### Abstract

*Air pollution is the contamination of air due to the presence of substances in the atmosphere that are harmful to the health of humans and other living things, or cause damage to the climate or to materials. The immune system is a complex defense network composed of cells, tissues, and organs that work together to protect the body against harmful substances. Several reports have revealed the direct association between exposure to poor air quality and increasing rate of morbidity and mortality mostly due to cardiovascular and respiratory diseases. Given the increasing levels of air pollution in urban and industrialized areas, understanding its effects on immune function is critical for public health. This paper explores how air pollution influences immune responses, including its impact on immune cell activation, inflammatory pathways, and susceptibility to diseases. Evidence shows that air pollutants can affect different immune cell types such as particle-clearing macrophages, inflammatory neutrophils, dendritic cells that orchestrate adaptive immune responses, and lymphocytes that enact those responses. A common theme that emerges is the capacity of air pollutants to stimulate pro-inflammatory immune responses across multiple classes of immune cells. Air pollution can enhance T helper lymphocyte type 2 (Th2) and T helper lymphocyte type 17 (Th17) adaptive immune responses, as seen in allergy and asthma, and dysregulate anti-viral immune responses.*

**Keywords:** Immune system, air pollution, air pollutants, effects.



## Reducing Environmental Hazards to Improve Public Health: Evidence from an Experimental Study in Nigerian Communities (A Case Study of Kwara and Oyo State)

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### Abstract

*Environmental hazards such as air and water pollution, poor waste management, and toxic chemical exposure remain critical contributors to the global burden of disease, disproportionately affecting low- and middle-income countries. Nigeria faces particularly high risks due to rapid urbanization, weak environmental regulation, and inadequate infrastructure. This study experimentally evaluated the impact of integrated environmental hazard reduction interventions on environmental quality and toxic biomarker levels in vulnerable communities in Kwara and Oyo States, Nigeria. A 12-month community-based experimental study was conducted across five urban, peri-urban, and rural communities (n = 750 households). Intervention households received improved biomass cook stoves, ceramic water filters, waste management tools, and environmental health education, while control households continued usual practices. Data collection at baseline, 6 months, and 12 months included environmental monitoring (PM<sub>2.5</sub>, E. coli counts) and biomarker analysis (blood lead, arsenic, and C-reactive protein). Statistical analysis employed chi-square, t-tests, and difference-in-differences models at a 5% significance level. After 12 months, the intervention group demonstrated significant improvements compared to controls: PM<sub>2.5</sub> levels declined by 38.4% ( $p < 0.001$ ), E. coli counts by 62.1% ( $p < 0.001$ ), and waste management scores improved by 74.3% ( $p < 0.001$ ). Biomarker analyses revealed reductions in blood lead (-54.7%), arsenic (-47.2%), and systemic inflammation (-36.5%) in intervention households ( $p < 0.001$ ). The findings provide robust experimental evidence that integrated environmental interventions can substantially improve environmental quality, reduce infectious disease burden, and lower toxic exposures in Nigerian communities. These results underscore the importance of incorporating environmental health interventions into national public health strategies and advancing multi-sectoral, community-based approaches to achieve Sustainable Development Goals.*

**Keywords:** Environmental hazards, Nigeria, Experimental study, air pollution, water quality, biomarkers.



## Field Evaluation of Tomato (*Solanum lycopersicum* L.) Genotypes for Resistance to Tomato Fruit Borer (*Helicoverpa armigera* Hubner) in Southwestern Nigeria

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### Abstract

*This appraisal of tomato genotypes against tomato fruit borer (*Helicoverpa armigera*) was conducted in replicated field trials during the late seasons of 2020 and 2021. The 20 tomato genotypes and accessions comprised: Alausa Long, Anaya F1, Ibadan Local, Kelvin F1, Mona F1, NGB00696, NGB00713, NGB00722, NGB00723, NGB00724, NGB00725, NGB00737, Panther-17, Rio Grande, Rodeo-62 F1, Roma Savanna, Roma VF (Susceptible check), Tima, Tropimech and UC82B. Anaya F1, Rio Grande, Mona F1, NGB00724 and UC82B had the lowest number of larvae/plant (0.04, 0.04, 0.05, 0.06, and 0.07) over the 2 years. Anaya F1, Mona F1 and NGB00725 showed the lowest mean percentage of fruit infestation (1.0, 4.7 and 13.1%), respectively. During the first year, Alausa Long, Mona F1 and Kelvin F1 had the highest fruit yields (30.7, 30.2 and 29.4 kg/ha, respectively), while Anaya F1, NGB00725 and Rio Grande had the highest yields (27.6, 16.4 and 15.9 kg/ha, respectively) during the second year. The correlations between pericarp thickness and both number of larvae and mean percentage of fruit damage ( $r = -0.827$ ,  $-0.802$ , respectively) were negative and significant. It was observed during the study that none of the genotypes was free from the *Helicoverpa armigera* attack. However, Anaya F1, Mona F1 and NGB00724 performed better with minimal *Helicoverpa armigera* larval infestation and had higher yields.*

**Keywords:** Tomato, *Helicoverpa armigera*, resistance, tomato genotypes, yield.





## Development of A New Lomax–Frechet Distribution: Properties, Applications and Model

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### Abstract

*The demand for flexible statistical models capable of capturing complex data behavior has led to the development of generalized distribution families. In this study, we introduce the New Lomax Frechet (NL–F) distribution, which integrates the heavy-tailed Frechet distribution within the New Lomax-G family. This new model aims to enhance flexibility in modeling skewed, heavy-tailed, or extreme-value data. We compare the NL–F distribution against three well-established models Lomax, Frechet, and Expo nential using a combination of statistical fit criteria and graphical assessments. The evaluation involves log-likelihood, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the Kolmogorov–Smirnov (K-S) statistic. Empirical results demonstrate that the NL F distribution achieves the highest log-likelihood and the lowest AIC, BIC, and K-S values, confirming its superior fit and efficiency. Visual comparisons via overlaid theoretical densities further substantiate these findings, showing that the NL F model most accurately captures the empirical distribution’s peak, spread, and tail behavior.*

**Key words and phrase:** Lomax distribution, Frechet distribution, New distribution, Data modeling, Statistical analysis, Risk analysis, Reliability.



## A Review on Side Effects of Analgesic on Humans

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### Abstract

*Pain is a complex and multidimensional experience that significantly affects the quality of life of individuals suffering from various medical conditions. It can be classified into different types, including nociceptive, neuropathic, and oncoplastic pain, each requiring distinct management strategies. Among the available treatments for pain relief, analgesics remain the most commonly used medications. These drugs help alleviate pain through various mechanisms, including the inhibition of pain signal transmission and modulation of pain perception at different levels of the nervous system. One of the major concerns in the use of analgesics is the occurrence of adverse effects. The side effects associated with opioid analgesics include tolerance, dependence, constipation, nausea, vomiting, respiratory depression, and cognitive impairment. Research has shown that the fear of analgesic side effects contributes to non-compliance among patients, particularly those undergoing cancer treatment, as they may opt for alternative pain management strategies such as acupuncture. Given the widespread use of analgesics and the substantial burden of their side effects, ongoing research and clinical studies are necessary to enhance the safety and effectiveness of pain management strategies. Hence, this review focused on analgesics, their side effects, and the prevention of these side effects. Future advancements in analgesic pharmacology, including the development of novel analgesics with fewer adverse effects, may offer promising solutions for improving patient outcomes.*

**Keywords:** pain, analgesic, adverse effects, medications.



## Human Face Detection: Advancements, Challenges, and Sustainable AI Perspectives

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### Abstract

*Human face detection has become a cornerstone of modern intelligent systems, underpinning applications in security, healthcare, human–computer interaction, and smart city infrastructures. Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML), particularly deep learning and transformer-based architectures, have revolutionized the accuracy and efficiency of face detection, enabling real-time deployment even on resource-constrained devices. This paper provides a comprehensive review of face detection methodologies, tracing the evolution from traditional handcrafted feature approaches to state-of-the-art deep learning models such as YOLOv7, RetinaFace, and Vision Transformers. Furthermore, it examines persistent challenges—including illumination variability, occlusion, and ethical concerns—and highlights strategies for fairness, transparency, and bias mitigation. By leveraging explainable AI, lightweight architectures, and fairness-aware training, the study emphasizes how intelligent face detection systems can contribute to sustainable development through secure digital ecosystems, enhanced public safety, and equitable access to technology. Ultimately, the integration of robust and responsible AI-driven face detection solutions has the potential to foster resilient, inclusive, and ethically governed smart societies.*

**Keywords:** Face Detection, Deep Learning, Vision Transformers, Explainable AI, Sustainable Development.



## Microbial Quality of a Restaurant Dishware and Dish Washing Water in Offa, Kwara State, Nigeria

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### Abstract

*Ensuring the microbiological safety of food and food-handling environments is paramount to preventing public health risks, particularly in communal settings. This study assessed the microbial quality of dishware and dishwashing water at a restaurant in Offa, Kwara State, Nigeria, to determine the level of contamination and identify the specific microorganisms present. Standard microbiological techniques, including cultural characteristics, Gram staining, and biochemical tests, were employed to isolate and identify the microbial flora. The results revealed significant contamination, with a Total Aerobic Plate Count (TAPC) of  $1.4 \times 10^3$  CFU/ml for dishwashing water and  $4.2 \times 10^3$  CFU/ml for dishware. The Total Fungi Count (TFC) was  $6.0 \times 10^3$  SPU/ml for dishwashing water and  $1.5 \times 10^3$  SPU/ml for dishware. The study identified several potentially pathogenic bacteria, including **Salmonella** species, **Escherichia coli**, **Campylobacter** species, and **Staphylococcus** species. Fungal isolates included **Aspergillus** species, **Mucor** species, and **Cladosporium** species. The presence of these microorganisms indicates a critical failure in hygienic practices and highlights a significant risk of food-borne illnesses and infections. These findings underscore the urgent need for improved sanitation and rigorous hygiene monitoring at this food-vending establishment to safeguard public health.*

**Keywords:** Microbial quality, Food safety, **Salmonella**, **Escherichia coli**, Food-borne pathogens



## Evaluation of Hygienic Status of Spoon and Dish Rinsing Water in a Restaurant of a Tertiary Institution in Offa, Kwara State, Nigeria

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### Abstract

*The microbiological quality of food service environments, particularly in shared spaces like institutional restaurants, is critical for public health. This study investigated the hygienic status of spoons and dish-rinsing water in a restaurant located within a tertiary institution in Offa, Kwara State, Nigeria. The objective was to determine the microbial load and identify the types of microorganisms present, using standard microbiological methods. Samples were collected, and standard cultural, staining, and biochemical tests were performed to isolate and identify the bacterial and fungal agents. The results revealed significant microbial contamination. The Total Aerobic Plate Count (TAPC) for spoons and dish-rinsing water were  $1.4 \times 10^3$  and  $4.2 \times 10^3$  colony-forming units per milliliter (CFU/ml), respectively. The Total Coliform Count (TCC) was  $7.6 \times 10^3$  CFU/ml for spoons and  $4.0 \times 10^3$  CFU/ml for dish-rinsing water. Additionally, the Total **Staphylococcus** Count (TSC) was  $1.5 \times 10^3$  CFU/ml for spoons and  $2.5 \times 10^3$  CFU/ml for dish-rinsing water, while the Total Fungi Count (TFC) was  $6.0 \times 10^3$  CFU/ml for spoons and  $1.5 \times 10^3$  CFU/ml for dish-rinsing water. The identified microorganisms included potentially pathogenic bacteria such as **Staphylococcus aureus**, **Shigella** species, and **Escherichia coli**, as well as fungi including **Epicoccum nigrum** and **Candida** species. These findings indicate a significant failure in hygiene practices and highlight the presence of hazardous microorganisms at this food-vending location. Urgent intervention is recommended to improve sanitation and prevent potential food-borne illnesses and infections among the student population.*

**Keywords:** Microbial contamination, Food hygiene, **Staphylococcus aureus**, Public health



## IoT-Enabled Home Security: Motion Detection with Automated User Notifications

By

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Between 9<sup>th</sup> – 11<sup>th</sup> September, 2025

### Abstract

*The growing demand for smart and affordable home security has accelerated the adoption of Internet of Things (IoT)-based solutions. This paper presents the design and implementation of an IoT-enabled home security system that integrates motion detection with automated user notifications. The system employs an Arduino Uno microcontroller as the primary controller, interfaced with a Passive Infrared (PIR) sensor to detect human movement and a buzzer to provide immediate audible alerts. An ESP8266 Wi-Fi module enables seamless connectivity to the Blynk IoT platform, facilitating real-time alerts delivered directly to users' smartphones. The software implementation, developed using Arduino IDE with C++, handles sensor monitoring, buzzer activation, and data communication. The system architecture was designed to combine local responses with remote monitoring, ensuring comprehensive protection. Experimental testing under varied conditions, including different lighting and movement scenarios, confirmed high detection accuracy, instant buzzer response, and notification delivery within an average of two seconds. The prototype demonstrated robustness, low latency, and scalability, making it suitable for practical residential applications. By integrating local and remote alert mechanisms, the system ensures dual-layered security, addressing the limitations of traditional alarm systems. The paper highlights the potential of IoT to provide reliable, cost-effective, and user-friendly security solutions. Future enhancements could include multi-sensor fusion, camera integration, and stronger encryption protocols to further improve performance, user trust, and system resilience.*

**Keywords:** Internet of Things (IoT), Home Security System, Motion Detection, Arduino Uno, User Notifications



## Microbiological Quality Assessment of Ready to Eat Food (Snacks) Sold Within the Mini Campus of the Federal Polytechnic, Offa, Kwara State Nigeria

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### Abstract

*The term ready-to-eat food means food that is ordinarily consumed in the same state as it is sold or distributed. Occasionally, human beings consume undesirable biological agents and toxins in the food leading to food-borne illness. This study was aimed at isolating and identifying the microorganisms from the ready-to-eat foods. The isolation of various microorganisms was carried out using selected media. Serial dilutions of the selected food samples were carried out aseptically and the inoculation of the sample was done using pour plate technique. Staining techniques and biochemical characterization were carried out on each of the isolates. The results obtained revealed bacteria such as Salmonella spp, Shigella, Pseudomonas, Enterobacter aerogenes Staphylococcus aureus. Fungi isolated include Mucor and Fusarium. The highest bacterial count of  $2.1 \times 10^2$  cfu/g was observed in egg roll sample while the least bacterial count was observed in chin chin with the value of  $1.8 \times 10^1$  cfu/g. The fungal isolates were found to be predominant in doughnut sample with value of  $2.7 \times 10^1$  cfu/g while the least fungal value was also found in chin chin with value of  $0.9 \times 10^1$  cfu/g. Untidy environment and poor personal hygiene are the major factors that prone the ready-to-eat foods as vehicles of disease transmission. Therefore, environmental sanitation and proper personal hygiene should be encouraged.*

**Keywords:** Ready-to-eat foods, bacteria, fungi, food borne illness.





## Uncovering the Health Risks of Artificial Sweeteners: A Review

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### Abstract

*Artificial sweeteners, widely used as low-calorie alternatives to sugar, have raised concerns about potential health risks. Specifically, this review examines the impact of artificial sweeteners, such as aspartame, sucralose, and acesulfame-K, on cardiovascular, hepatic, and renal health. Our analysis of existing literature reveals potential associations between artificial sweetener consumption and increased risk of cardiovascular disease, liver damage, and kidney problems. Furthermore, underlying mechanisms include alterations in gut microbiota, insulin resistance, and increased oxidative stress and inflammation. These findings highlight the need for further research to inform public health policy. Additionally, they underscore the importance of raising awareness among consumers and policymakers about potential health risks. Ultimately, this review aims to contribute to the ongoing discussion on the safety of artificial sweeteners and encourage informed decision-making.*

**Keywords:** *Artificial sweeteners, Health risks, Public health policy*



## AI Model for Inflation Prediction: A Macroeconomic Indicator-Based Approach in Nigeria

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### Abstract

*This paper presents the design and implementation of an Artificial Intelligence (AI)-based model for forecasting inflation in the Nigerian economy using macroeconomic indicators, with a view to providing an accurate, data-driven tool to support evidence-based policymaking, enhance economic planning, and contribute to achieving greater macroeconomic stability. Monthly macroeconomic data from 1995 to 2024 were sourced from the Central Bank of Nigeria and the National Bureau of Statistics websites. Four models, Autoregressive Integrated Moving Average (ARIMA), Random Forest, Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM) neural networks, were developed and evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Results show that AI-based models consistently outperformed ARIMA, with the LSTM model achieving the best performance. The findings demonstrate the superiority of AI approaches in capturing nonlinear and temporal dependencies in Nigeria's inflation dynamics. This research contributes to the development of evidence-based forecasting tools that can enhance monetary policy, economic planning, and financial decision-making in Nigeria. Future work should focus on integrating high-frequency indicators, incorporating explainable AI techniques, and deploying real-time dashboards for policymakers.*

**Keywords:** Inflation Forecasting, Artificial Intelligence, Machine Learning, Nigerian Economy, ARIMA, Economic Policy.



## Investigation of Structural and Electronic Properties of InN and Its Alloys (InN Doped with Al) Using Density Functional Theory

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### Abstract

*This research was carried out to investigate the structural and electronic of pure InN and  $In_{1-x}Al_xN$  ( $x= 0.25, 0.5, 0.75$  and  $1$ ) using Density Functional Theory (DFT). The full potential linear augmented plane wave (FP-LAPW) was used to perform the calculations based on DFT as executed in the WIEN2K software. The two exchange correlational function employ in this research are gradient approximation (GGA) and TB-mBJ. The research was carried out on simple cubic InN sample and the calculated lattice parameter for the zinc blende pure InN is  $a=b=c= 5.06 \text{ \AA}$ . The parameters of all the doped samples were influenced by the dopant and the percentage of doping used. The electronic properties of the samples were investigated and the samples were found to have a stupendous application due to the changes in the density of state influencing the band gap when the Al concentration was increased. A direct band gap and a semiconductor nature were confirmed for both pure InN and Al doped InN ( $In_{1-x}Al_xN$ ;  $x= 0.25, 0.5, 0.75$  and  $1$ ). The underestimated band gap of PBE was improved by the TB-mBJ potential, which showed a far better performance than the GGA-PBE potential.*

**Keywords:** Alloy, dope, DFT, FP-LAPW, Structural and Electronic Properties.



## Insecticidal Assessment of Oystershell Ash and Cowhorn Ash Against *Sitophilus Zeamais* Motschulsky in Stored Maize

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### Abstract

*Maize (Zea mays L.) is an essential staple crop globally but faces postharvest losses due to Sitophilus zeamais, which reduces grain quality and viability. While synthetic pesticides are effective, their environmental and health concerns necessitate sustainable alternatives. This study assessed the efficacy of Oyster Shell Ash (OSA) and Cow Horn Ash (CHA) in controlling S. zeamais infestation in stored maize. Bioassays were conducted using OSA and CHA at 2.0, 4.0, and 6.0 g/100 g maize, with Permethrin (0.6% w/w) and untreated grains as controls. Mortality was recorded at 3, 5, and 7 days post-treatment (DPT), while progeny emergence and kernel damage were evaluated at 40 DPT. OSA demonstrated superior insecticidal activity, achieving 88.75% mortality at 2.0 g and 97.50% at 6.0 g/100 g maize by 7 DPT. CHA was less effective, with 86.25% mortality at 6.0 g/100 g maize, while Permethrin reached 100% mortality at 7 DPT. OSA completely inhibited progeny emergence at its highest dose, whereas CHA allowed limited development. OSA-treated grains had 3.18% kernel damage at 40 DPT, lower than CHA (4.50%), while untreated grains had significantly higher (16.12%) damage ( $p=0.05$ ). Seed germination remained unaffected, with OSA- and CHA-treated seeds maintaining 91.25–93.75% viability. These findings highlight OSA as a promising alternative to synthetic insecticides. Further research is needed to optimize their long-term effectiveness in maize storage.*

**Keywords:** Maize, *Sitophilus zeamais*, percentage mortality, weight loss, grain damage, Oystershell Ash, and Cowhorn Ash.



## Development of A New Lomax–Frechet Distribution: Properties, Applications and Model

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### Abstract

*The demand for flexible statistical models capable of capturing complex data behavior has led to the development of generalized distribution families. In this study, we introduce the New Lomax Frechet (NL–F) distribution, which integrates the heavy-tailed Frechet distribution within the New Lomax-G family. This new model aims to enhance flexibility in modeling skewed, heavy-tailed, or extreme-value data. We compare the NL–F distribution against three well-established models Lomax, Frechet, and Expo nential using a combination of statistical fit criteria and graphical assessments. The evaluation involves log-likelihood, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the Kolmogorov–Smirnov (K-S) statistic. Empirical results demonstrate that the NL F distribution achieves the highest log-likelihood and the lowest AIC, BIC, and K-S values, confirming its superior fit and efficiency. Visual comparisons via overlaid theoretical densities further substantiate these findings, showing that the NL F model most accurately captures the empirical distribution’s peak, spread, and tail behavior.*

**Keywords:** Lomax distribution, Frechet distribution, New distribution, Data modeling, Statistical analysis, Risk analysis, Reliability.



## Polytechnic Libraries as Agents of Innovation and Entrepreneurship in Ogun State

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### Abstract

*This study examined the role of polytechnic libraries in Ogun State as agents of innovation and entrepreneurship. The objectives were to identify the resources and services available to support innovation, assess the extent of their utilization, evaluate their perceived effectiveness, and identify challenges faced. The study adopted a descriptive survey design, using questionnaires administered to library personnel across selected polytechnics in Ogun State. Data were analyzed using descriptive and inferential statistics, while interpretations were guided by the Resource-Based View (RBV) theory, Diffusion of Innovation theory, and the Innovation Ecosystem framework. Findings revealed that polytechnic libraries possess basic print and electronic collections, ICT tools, and some digital services, but lack specialized entrepreneurial resources such as market intelligence databases, makerspaces, and innovation labs. Utilization of entrepreneurship-related resources was moderate, with many students relying more on open-access digital content than on services provided by the library. Libraries were perceived as moderately effective in promoting innovation and entrepreneurship, but their impact was hindered by limited funding, inadequate ICT infrastructure, erratic power supply, and insufficient professional training of librarians. The study concluded that polytechnic libraries in Ogun State hold significant but underutilized potential as agents of innovation and entrepreneurship. The study recommended a multi-stakeholder approach involving government, institutions, and industry to reposition polytechnic libraries as dynamic partners in fostering entrepreneurship and sustainable development.*

**Keywords:** Diffusion of innovation, Entrepreneurship, Innovation ecosystem, Innovation, Polytechnic libraries, Resource-based view.



## Advanced Sensor and Microcontroller-Based Approach for Smart Irrigation Systems

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### Abstract

*This study presents the design and development of an advanced sensor and microcontroller-based smart irrigation system that integrates soil moisture sensing and automated water control. Using an Arduino Uno, a relay module, and a soil moisture sensor, the system monitors real-time soil conditions and activates irrigation only when required. The Arduino Uno R3 code sets up the soil moisture sensor and the water pump, and continuously monitors the soil moisture level. A threshold value of 900 was set on a scale of 0 to 1023 for the soil moisture sensor, where values below 900 indicate dry soil to trigger the water pump. Experimental results demonstrate reliable control of water delivery, with moisture threshold calibration enabling optimization for different crops and soils. The proposed system minimizes water waste, reduces labour costs, and enhances sustainable farming practices, illustrating the potential of precision agriculture technologies in resource-constrained environments.*

**Keywords:** *Sensors, Microcontroller, Smart irrigation system, and Sustainable farming.*





## Trends, Challenges, and Novel Strategies in Combating Antimicrobial Resistance: A Comprehensive Review

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### Abstract

*This seminal paper provides a comprehensive overview of antimicrobial resistance (AMR), its historical evolution, and strategies for combating it. The review synthesizes information on the emergence of resistance in microorganisms, highlighting the role of both natural mutations and human-driven factors like the misuse of antibiotics. It explores the mechanisms of resistance, such as horizontal gene transfers and efflux pumps, and discusses novel approaches to counter these challenges. Key strategies highlighted include the development of antimicrobial peptides (AMPs), which exhibit broad-spectrum activity with minimal resistance development, and the use of enzyme and efflux pump inhibitors as antibiotic adjuvants. The paper emphasizes the urgency of addressing AMR as a silent pandemic threatening global health and food security, and it underscores the need for increased surveillance and responsible antibiotic stewardship.*

**Keywords:** Antimicrobial resistance, superbugs, antibiotic stewardship, antimicrobial peptides, efflux pumps.



## Epidemiology and Risk Factor of Nosocomial Infections in Different Setting (Intensive Care Unit, Neonatal Unit, And Surgical Unit)

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### Abstract

*Nosocomial infections (NIs), also known as healthcare-associated infections, are infections acquired 48 hours or more after hospital admission without prior evidence of infection. They represent a critical global health challenge, significantly contributing to morbidity, mortality, prolonged hospital stays, and increased healthcare costs. Common types include catheter-associated urinary tract infections, central line-associated bloodstream infections, surgical site infections, and ventilator-associated pneumonia. This paper examines the epidemiology and risk factors of nosocomial infections across neonatal units, surgical wards, and intensive care units, with emphasis on developing countries where the burden is highest due to inadequate infrastructure, poor infection control practices, and antimicrobial resistance. Prematurity, invasive medical procedures, prolonged hospitalization, overcrowding, and compromised immunity were identified as major determinants of infection risk. The review highlights that nosocomial infections account for 4–56% of neonatal deaths in sub-Saharan Africa, with prevalence rates in low- and middle-income countries reaching up to three times that of developed regions. Despite global efforts, antimicrobial resistance exacerbates the problem, limiting treatment options. Effective control requires comprehensive infection prevention strategies, including strict adherence to hygiene protocols, antimicrobial stewardship, proper hospital waste management, surveillance systems, and continuous education of healthcare workers and patients. By addressing modifiable risk factors and strengthening infection control programs, healthcare facilities, especially in resource-limited settings, can reduce the burden of nosocomial infections and improve patient outcomes.*

**Keywords:** Nosocomial infections, Risk factors, Antimicrobial resistance, Infection control.



## Antimicrobial Profile and Phytochemical Screening of *Hunteria Umbellate* (Abere) Seeds Extract Against Selected Microorganisms

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### Abstract

*Hunteria umbellata* locally known as **Abere**, is a medicinal plant widely used in West Africa for the management of infectious and metabolic diseases. This study evaluated the antimicrobial activity and phytochemical composition of the ethanolic and petroleum ether seed extracts of *H. umbellata* against selected pathogens. Antimicrobial screening was conducted at concentrations of 100, 50, and 25 mg/mL using the agar well diffusion method, while Minimum Inhibitory Concentration (MIC) was determined by broth dilution. Phytochemical constituents were assessed qualitatively. The results showed that both extracts inhibited bacterial growth in a concentration-dependent manner, with ethanolic extracts producing larger inhibition zones than petroleum ether extracts. At 100 mg/mL, the ethanolic extract showed the highest activity against *Escherichia coli* (22.63 mm), while petroleum ether extract was most effective against *Bacillus subtilis* (19.56 mm). Phytochemical analysis revealed the presence of tannins, flavonoids, alkaloids and saponin. These findings are consistent with recent reports that highlight the solvent-dependent antibacterial potential of *H. umbellata*. In conclusion, the seeds of *H. umbellata* possess significant antimicrobial activity, particularly in ethanolic extracts, and their phytochemical richness supports their ethnomedicinal relevance. Further studies are recommended.

**Keywords:** Ethanolic, Petroleum ether, Abere, Extraction.



## Analysis of Digital Transformation in Modern Libraries: Technologies, Challenges, and Future Directions

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### Abstract

*The information age has catalyzed a fundamental paradigm shift, transforming libraries from traditional knowledge repositories into dynamic, technology-driven information hubs. This analysis examines the multifaceted process of digital transformation within modern libraries, focusing on the integration of disruptive technologies such as Artificial Intelligence (AI), big data analytics, cloud computing, and the Internet of Things (IoT). These innovations enable personalized user services, data-driven resource management, and the creation of smart library spaces, fundamentally altering service models and operational efficiency. However, this transformation presents significant challenges, including financial constraints, organizational resistance, and the risk of exacerbating the digital divide. Success hinges on strategic implementation through comprehensive digital strategies, maturity model assessment, strategic partnerships, and a steadfast commitment to user-centered design. This paper concludes that while the journey is complex, navigating these technological, organizational, and strategic dimensions is essential for libraries to evolve from passive repositories into active participants in the knowledge creation process and remain vital, relevant institutions in the digital era.*

**Keywords:** Digital Transformation; Smart Libraries; Technical Challenges; Future Directions.



## Traffic Flow Dynamics at Offa Garage Roundabout in Ilorin, Kwara: Implications for Urban Management

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### Abstract

*The rapid urbanization of Ilorin, Nigeria, has resulted in increased traffic congestion at critical intersections, particularly at the Offa Garage Roundabout, which serves as a key junction linking residential, commercial, and transit corridors. This research intends to evaluate the traffic flow at this roundabout to identify operational inefficiencies and develop viable mitigation strategies. The research objectives include assessing geometric features, determining traffic capacity, and analysing the Level of Service (LOS). Field data were collected through manual traffic counts, speed measurements, and delay studies conducted during both peak and off-peak hours. The analysis examined geometric parameters, including entry and exit widths, the diameter of the inscribed circle, and circulatory lane configurations, alongside traffic volume and composition. The findings revealed significant operational challenges: peak hour traffic volumes exceeded 1,200 vehicles per hour, average delays reached 45 seconds per vehicle, and queue lengths extended up to 150 meters. The roundabout is currently operating at LOS D, indicating a state of high congestion and reduced efficiency. Contributing factors to these issues include a heterogeneous traffic mix (60% cars, 20% motorcycles, 10% buses, and 10% trucks), inadequate geometric design, and non-compliance with traffic regulations. Recommendations for improvement include widening entry lanes, optimizing the circulatory roadway, reconfiguring the central island, and enhancing traffic enforcement and public awareness initiatives. These interventions are intended to improve traffic flow, minimize delays, and enhance safety. This study underscores the necessity of addressing urban mobility challenges in rapidly growing cities like Ilorin, providing actionable insights for policymakers and urban planners aimed at fostering sustainable transportation systems.*

**Keywords:** Congestion mitigation, Geometric design, Level of Service (LOS), Roundabout efficiency, Traffic flow, Urban mobility.



## Artificial Intelligence in Education: Balancing Innovation and Ethics for Sustainable Learning

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### Abstract

*Artificial Intelligence (AI) is transforming the educational landscape by introducing intelligent solutions that enhance learning outcomes, optimize resource utilization, and promote inclusivity. This paper examines three key applications—AI-powered chatbots, personalized learning systems, and automated assessment tools—that collectively redefine how education is delivered and accessed. These innovations not only improve teaching efficiency and learner engagement but also contribute to the broader goals of sustainable development by enabling scalable, affordable, and equitable education for diverse populations. However, the integration of AI in education raises critical ethical and operational challenges, including data privacy, algorithmic bias, and the potential for over-dependence on technology. Through practical examples and case studies, this paper provides a comprehensive analysis of these opportunities and risks, offering insights for educators, policymakers, and technologists committed to building resilient and sustainable educational ecosystems.*

**Keywords:** AI in education, personalized learning, automated grading, sustainable development, ethical AI.



## Cyber Threat Intelligence: Enhancing Proactive Security Measures on Infrastructure

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### Abstract

*Cyber Threat Intelligence (CTI) plays a critical role in modern cybersecurity by providing organizations with actionable insights into emerging threats, enabling them to identify and mitigate risks before they materialize. Cyber-attacks are significant threats affecting any organization, particularly Critical Infrastructure (CI) organizations; the health sector, autonomous vehicles, and the power sector. These attacks are increasingly sophisticated, multi-vector, and less predictable, making Cybersecurity Risk Management (CSRM) more challenging. To effectively counter these evolving threats, CI organizations require a robust security defense strategy that leverages CTI to minimize risk and enhance protection. CTI provides evidence-based information on threats, enabling organizations to adopt proactive security measures. These measures include real-time threat monitoring, predictive analytics, automated threat detection, and intelligence-driven incident response. Additionally, CTI facilitates the integration of machine learning, artificial intelligence (AI), open-source intelligence (OSINT), and threat intelligence sharing to enhance security posture. By leveraging these technologies, organizations can anticipate potential attacks, identify vulnerabilities, and implement preemptive security controls, ultimately strengthening their cybersecurity defenses. This seminar report examines the significance of CTI in cybersecurity, focusing on its role in threat identification, analysis, and dissemination. It explores methods, challenges, and benefits associated with CTI implementation, highlighting its effectiveness in proactive security strategies. Furthermore, the report delves into emerging trends such as context-aware intelligence and AI-driven threat detection, which offer promising advancements in improving cybersecurity resilience.*

**Keywords:** Cybersecurity, Cyber Threat Intelligence, Cybersecurity Risk Management, Critical Infrastructure, Security Controls, Threat Intelligence Sharing.





## Effect of Soyabean Replacement with Moringa - Cassava Leaves on the Sensory Evaluation and Serum Biochemical Profile of Funaab Alpha Broiler-Chicken

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### Abstract

*Soybean is a commonly used ingredient in poultry feed formulation due to its high protein content. However, the increasing competition between humans and livestock for soybean has led to price hikes, prompting the search for cost-effective alternative feed sources. This study aimed to assess the impact of replacing soybean with Moringa and Cassava leaf meal (MCLM) on sensory evaluation, and serum biochemical profile of FUNAABALPHA broiler-chicken. Five treatment diets were formulated, with MCLM replacing soybean at levels of 0%, 5%, 10%, 15%, and 20%. The study followed a completely randomized design with three replicates, each containing 15 birds, and lasted for 8 weeks. The results revealed that MCLM-based diets significantly influenced ( $P < 0.05$ ) the juiciness and color of the meat. Serum analysis revealed that the total protein, albumin and globulin values were not significantly influenced ( $P > 0.05$ ) by the treatments. Cholesterol values were higher ( $P < 0.05$ ) in T1 than in other diet. Calcium, potassium, sodium, phosphorus and bicarbonate levels were not influenced ( $P > 0.05$ ) by the treatments. Overall, the study suggests that replacing soya meal with MCLM in broiler diets can positively impact meat quality, and overall bird health, with potential hypocholesterolemic effects and maintenance of metabolic functions.*

**Keywords:** Soyabean, Moringa, cassava, Brioler chicken, Biochemical profile.



## The Impact of COVID-19 on Small Scale Business Enterprises in Kaduna

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### Abstract

*This research seeks to add to the discussions in academia and policy domain by studying the economic challenges faced by small scale business enterprises as a result of the Covid-19 pandemic in Kaduna State. Using business establishment frame from Kaduna State Bureau of Statistics, A sample of 360 small businesses were selected within the urban LGAs Kaduna North, Kaduna South, Chikun, Igabi, Zaria Sabon/gari and Jemaa. This study used a quantitative approach of data collection and data from the interviews was analyzed using SPSS package to generate table and charts. Of the 360 small businesses was randomly sampled and interviewed, 348 were successfully covered showing a response rate of 96.7%. The finding showed that 69.5% of businesses are registered with cooperate affairs commission. 30% of registered. The result showed that there is sizable decrease in business operation, production level, and quantity of sales and reduced investment opportunities due to the effect of COVID-19. Decrease in employee turnover. Only 53% of businesses are aware of any government's intervention efforts to support businesses in Kaduna. However only 2.3% benefited from any government intervention effort aimed at supporting businesses in Kaduna. Just as experienced in several other, the case in Kaduna state was not different as a lot of business suffer the as a result of the lockdown. There is need to create more awareness of any government interventions and as well as create policies that will improve the overall business activities in the state in times as experienced during the pandemic.*



## Comparative Correlation Analysis Among Radioactivity Levels of $^{226}\text{Ra}$ , $^{40}\text{K}$ and $^{232}\text{Th}$ in Selected Areas of Kaduna Methopolis, Kaduna State and Federal Polytechnic Kaltungo, Gombe State

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### Abstract

*The concentration levels of Radon ( $^{226}\text{Ra}$ ), Thorium ( $^{232}\text{Th}$ ) and Potassium ( $^{40}\text{K}$ ) in soil have a great concern in the recent decades, due to its effect on the human health. This research aimed at comparing relationships between detected naturally existing radioactive elements in selected locations in Kaduna state and Gombe state. Data was obtained from previously published research conducted in 2017 and 2023. Correlation analysis was conducted to determine the strength of association amongst elements. The result conducted shows that  $^{232}\text{Th}$  and  $^{40}\text{K}$  have a weak but significant positive relationship of 0.415 in Kaduna and in Gombe, only the relationship between  $^{40}\text{K}$  and  $^{226}\text{Ra}$  was significantly positive with a value of 0.539, Suggesting a positive, but weak relationship between the two elements. Given the results obtained from this analysis, it is safe to say that there is a high likelihood that there is a connection between  $^{40}\text{K}$  and  $^{226}\text{Ra}$  and  $^{40}\text{K}$  and  $^{232}\text{Th}$  if established would help environmentalist in the fight against the rise of these metals. It is our recommendation that more research be undertaking in line with this area so as to further affirm/criticize findings from this analysis.*

**Keywords:** Correlation, NaI(Tl) detector, radioactivity, Radon equivalent activity.



## Production and Quality Assessment of Muffin Cakes from Wheat Flour, Flax Seeds and Tigernut Flour Blends

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### Abstract

*Recently, consumers have shown increased interest in food ingredients due to health concerns such as celiac disease associated with wheat consumption and the economic implications of wheat dependence. Flax seeds and tigernut, known for their nutritional and health benefits, present opportunities for producing nutritionally enhanced muffins. This study investigated the effects of substituting wheat flour with flax seeds (*Linum usitatissimum*) and tigernut (*Cyperus esculentus*) on the quality attributes of muffins. Four formulations were prepared: sample A (100% wheat flour), sample B (90% wheat + 5% tigernut + 5% flax seed), sample C (80% wheat + 10% tigernut + 10% flax seed), and sample D (70% wheat + 15% tigernut + 15% flax seed). Standard methods were used to evaluate proximate, mineral, physical and sensory properties of muffins. Proximate results showed moisture (9.17–10.30%), crude protein (12.08–16.01%), crude fibre (0.86–1.96%), crude fat (13.99–22.30%), ash (2.14–2.73%), and carbohydrate (57.02–70.95%). Mineral contents ranged from 0.32–0.58 mg/100 g (iron), 13.11–15.77 mg/100 g (magnesium), and 0.24–0.48 mg/100 g (zinc). Physical properties showed muffin weights (27.99–51.02 g), heights (2.10–2.70 cm), volumes (76.18–138.92 cm<sup>3</sup>), and densities (0.21–1.11 g/cm<sup>3</sup>). Sensory evaluation indicated sample A (100% wheat) was most preferred, though composite samples were also acceptable. Sample D exhibited superior nutritional properties but lower carbohydrate content. The study concludes that acceptable, nutrient-rich muffins can be developed using wheat–tigernut–flaxseed blends.*

**Keywords:** Composite flour, Functional properties, Proximate composition, Mineral and Sensory evaluation.



## Artificial Intelligence Driven Solutions for Food Security and Post-Harvest Loss Reduction in Nigeria

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### Abstract

*Post-harvest losses continue to pose a significant threat to food security in Nigeria, with estimates suggesting that 30–50% of staple crop production is lost annually due to poor handling, inadequate infrastructure, and inefficient market systems. Addressing this challenge is critical to achieving national food security targets and supporting smallholder farmers who dominate Nigeria’s agricultural sector. This study applies Artificial Intelligence (AI) and Machine Learning (ML) techniques to identify and quantify the key drivers of post-harvest losses using a nationally representative dataset. A Random Forest regression model was developed and trained on 117 selected survey-based indicators, achieving strong predictive performance ( $R^2 = 0.87$ ,  $MAE = 0.27$ ,  $RMSE = 0.46$ ). Model interpretability was enhanced using SHAP (SHapley Additive exPlanations) values, which highlighted post-harvest storage conditions, availability of drying facilities, market access distance, and training exposure as the most influential predictors of crop loss rates. Beyond prediction, this research provides actionable insights for policymakers and development partners. Recommendations include upgrading rural storage and processing infrastructure, reducing farmers’ distance to markets through improved road networks and aggregation centers, and deploying farmer-centered capacity-building programs to minimize avoidable losses. The integration of AI-powered decision-support tools into agricultural extension services is also proposed to enable proactive loss prevention. The results demonstrate the transformative potential of data-driven approaches for optimizing resource allocation, guiding targeted interventions, and accelerating Nigeria’s progress toward Sustainable Development Goal 2 (Zero Hunger).*

**Keywords:** Crop Yield Prediction, Machine Learning, SHAP Analysis, Random Forest, Agricultural Data, Feature Importance, Predictive Analytics, Food Security, Policy Recommendations, Nigeria.



## The Effect of Data Privacy Regulations on Nigerian Insurer Innovation

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### Abstract

*Data privacy regulations have become increasingly important in shaping business innovation across industries globally. This study empirically investigates the effect of data privacy regulations on insurer innovation in Nigeria, focusing on the specific dimensions of regulatory compliance burden, data governance requirements, innovation constraints, and competitive positioning. Drawing on the Resource-Based View Theory, Innovation Diffusion Theory, and Institutional Theory, the research posits that these data privacy regulation dimensions significantly impact insurer innovation capabilities. A survey research design was employed, collecting data from 287 insurance executives, IT managers, and innovation specialists operating in Nigerian insurance companies. Data were analyzed using descriptive statistics, Pearson Product-Moment Correlation, and Multiple Regression Analysis. The findings reveal that all four independent variables, regulatory compliance burden, data governance requirements, innovation constraints, and competitive positioning, are significant predictors of insurer innovation. Specifically, competitive positioning emerged as the strongest positive predictor of insurer innovation ( $\beta=0.412$ ,  $p<0.001$ ), while regulatory compliance burden showed the strongest negative impact ( $\beta=-0.389$ ,  $p<0.001$ ). The regression model explained a substantial portion of the variance in insurer innovation ( $R^2=0.698$ ). This study concludes that while data privacy regulations create compliance challenges that can constrain innovation in Nigerian insurance companies, the strategic positioning of these regulations as competitive advantages and the development of robust data governance frameworks can enhance innovation capabilities. The results reinforce the theoretical frameworks, suggesting that strategic adaptation to data privacy regulations can transform regulatory requirements into innovation catalysts. Recommendations include developing integrated compliance-innovation strategies, investing in data governance infrastructure, and leveraging regulatory compliance as a competitive differentiator.*

**Keywords:** Data Privacy Regulations, Insurer Innovation, Nigerian Insurance Market, Regulatory Compliance, Data Governance, Innovation Constraints, Competitive Positioning.





## Design of A Framework for Network Fault Detection Using Java Mobile Agents

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### Abstract

*Fault detection plays a critical role in the broader process of fault diagnosis within computer networks. It generally involves two main stages, fault detection and fault localization. This study explores the use of Java multi-agent system, leveraging autonomous software agents that collaborate to monitor, detect, and diagnose network issues. The proposed system is designed to identify whether individual nodes are active (up) or inactive (down) and returns a Boolean status along with supplementary information about each visited node. The research presents the development of a fault detection framework based on java multi-agents architecture tailored for local area networks (LANs). The framework is composed of several software agents, including a CoordinatorAgent, MonitoringAgent, AnomalyDetector, NetworkAgent, Status Checker, and a backend database. The CoordinatorAgent is responsible for assigning tasks to the MonitoringAgent and AnomalyDetector behavior using unique identification numbers. The MonitoringAgent manages the actions of NetworkAgent and AnomalyDetector, watching their migrations across network nodes to perform probing and status reporting. The system was implemented using the JADEAgentCommunicator and deployed on a test network comprising ten nodes. Testing was conducted over a period of ten days, with the system operating for six hours each day. The experimental results was used to get the MTTD and MTTR and the accuracy of fault detection reliability ranging from 65% to 100%, indicating the system's potential effectiveness in real-world network environments. This research contributes to the field of autonomous fault detection in computer networks and provides a foundation for enhancing network reliability. Future work will aim to extend the framework to support full fault diagnosis, including the identification and classification of the root causes of node failures using advanced mobile agent techniques.*

**Keywords:** Mobile Agent, System Reliability, Network, JADE, Fault detection, MTTD, MTTR





## Phytochemical Screening and Antibacterial Analysis of *Morinda Lucida* Fruit

### Extract on Some Bacterial Isolates

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### Abstract

The phytochemical and antibacterial activity of Methanolic extract of *Morinda lucida* fruit was studied. The phytochemical test was carried out employing standard procedures. Antibacterial activity of the extract was detected against some bacterial pathogens. The Fruit extract was screened for antibacterial activities at a concentration of 100mg/ml. The antibacterial efficacy of the extract against the bacterial isolates was evaluated by the appearance of clear zones of inhibition around the wells. Methanolic extract of *Morinda lucida* fruit exhibited antibacterial activity against *Haemolytic streptococcus*, *Salmonella* spp, *Pseudomonas* spp, *Staphylococcus aureus*, *Proteus* spp, *Acinetobacter* spp, *Escherichia coli*, *Klebsiella* spp and *Streptococcus faecalis*. Methanolic extract of *Morinda lucida* fruit showed strong antibacterial activity against *Haemolytic streptococcus*, *Pseudomonas* spp, *Proteus* spp, *Acinetobacter* spp and *Streptococcus faecalis* at the concentration of 100mg/ml with zones of inhibition ranging between 3mm and 22mm. Phytochemical analysis of active extract demonstrated the presence of common phytoconstituents like Tannins, Glycosides, Saponins and Flavonoids. The percentage yield for methanol extract of *Morinda lucida* fruit is 4.8%. Results from the antibacterial activity analysis revealed that the methanolic extract of *Morinda lucida* fruit at different concentrations were more active compared with standard antibiotics against all the tested isolates used. The Minimal Inhibitory Concentrations for all tested isolates were at concentrations ranging from 12.5mg/ml to 100mg/ml. These phytochemical established a good support to the use of this plant in herbal medicine and as good candidate for the development of new drugs.

**Keywords:** *Morinda lucida*, Antibacterial screening, Methanolic extract, Phytochemical screening, sensitivity test.



## Characterization and Processing of Ikere - Ekiti Clay for Industrial Application

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### Abstract

*In this research, kaolin was sourced from the kaolin deposit located in Ikere-Ekiti, Ekiti State. The pretreatment was done by soaking the clay- in water for three days followed by decantation. The prepared clay was characterized using ED-XRF to determine its elemental composition, this was followed by sieve analysis using mesh size 63 $\mu$ m. The clay was calcined and processed using scrubbing method followed by characterization of the clay concentrate and tailing. The ED-XRF result shows that the kaolin clay contains 38.45 %  $Al_2O_3$ , 55.89 %  $SiO_2$ , 0.49 %  $K_2O$ , 1.56 %  $TiO_2$ , 2.44 %  $Fe_2O_3$  and 0.31 %  $ZrO_2$  while the concentrate of the processed kaolin contain 51.70 %  $Al_2O_3$ , 43.86 %  $SiO_2$ , 1.39 %  $TiO_2$ , 0.78 %  $Fe_2O_3$  and 1.04 %  $ZrO_2$  and lastly the tailing of the processed kaolin contain 13.35 %  $Al_2O_3$ , 73.01 %  $SiO_2$ , 3.32 %  $K_2O$ , 1.42 %  $TiO_2$ , 5.41 %  $Fe_2O_3$  and 0.03 %  $ZrO_2$ . It was concluded that the Ikere-Ekiti kaolin after been upgraded through calcination and scrubbing process was found suitable for industrial uses.*



## Microbial Analysis of Some Selected Sachet Water in Ilorin Metropolitan

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### Abstract

*Access to clean drinking water is vital for public health, especially in developing regions where sachet water is commonly consumed. The physicochemical and microbiological quality of five sachet water brands ABS Water (AW), Basic Water (BW), Kwara Poly Water (KW), Olawoyin Water (OW), and Yinto Wtaer (YW) produced in Ilorin, Kwara State, Nigeria were evaluated. Parameters such as pH, temperature, turbidity, total dissolved solids (TDS), electrical conductivity (EC), dissolved oxygen (DO), and salinity were analyzed using standard methods. Microbiological assessment included total viable count (TVC), total coliforms count (TCC), Total **Escherichia coli** (TEC), and Total fungi Count (TFC), with bacterial isolates identified through morphological and biochemical tests. Results showed that all brands had acceptable turbidity (1.60–4.20 NTU), temperature (18.00–24.00°C), EC (51.00–100.00  $\mu$ S/cm), and TDS (35.00–59.00 mg/L) levels within WHO guidelines. DO ranged from 3.20 mg/L in AW to 17.30 mg/L in OW. However, microbiological results revealed concerns: YW recorded the highest TVC (231.00 CFU/mL), exceeding permissible limits, while OW had the lowest (7.00 CFU/mL). Total coliforms were found in four brands, ranging from 3.00 to 51.00 CFU/mL. **E. coli** was absent in all samples. Identified bacterial isolates included **Bacillus subtilis**, **Staphylococcus epidermidis**, **Pseudomonas aeruginosa**, and **Neisseria spp.**, while **Aspergillus spp.** was found in YW. Although some sachet water brands complied with WHO standards, others posed health risks due to microbial contamination. The study recommends stricter regulation, improved hygiene in production, and continuous monitoring to ensure water safety for consumers.*

**Keywords:** drinking water quality, microbial contamination, physicochemical parameters, public health.



## Synthesis and Characterization of Bioethanol from Pineapple Crown

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### Abstract

*This research investigates the production of bioethanol from African pineapple crown. The process involved several steps, including sample collection, pretreatment with H<sub>2</sub>SO<sub>4</sub>, fermentation using **Saccharomyces cerevisiae**, and distillation. The physicochemical properties of the produced bioethanol of light brown coloration and unpleasant odour were evaluated. Key findings include an iodine value of (14 g I<sub>2</sub>/100g), acid value of (29.45 mg KOH/g), and a refractive index of (1.3601). Other properties such as density (1.205 g/mL), specific gravity (0.987g/cm<sup>3</sup>), cloud point (8°C), and a 60% yield were also recorded. Fourier-transform infrared spectroscopy (FTIR) analysis confirmed the presence of functional groups typically found in bioethanol, including O-H(3327.41cm<sup>-1</sup>), C=O(1633.48cm<sup>-1</sup>), and C-O(1049.75cm<sup>-1</sup>) stretchings. The results suggest that pineapple crown could serve as a viable source for bioethanol production, providing an eco-friendly and renewable energy alternative. However, further purification is recommended to improve bioethanol quality and meet industry standards.*



## Synthesis and Characterization of Bio-Ethanol from Paper Waste

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### Abstract

*This study investigates the production of bioethanol from paper waste using acid pretreatment, fermentation, and distillation. Paper waste collected from the Federal Polytechnic Offa, Kwara State, Nigeria, was treated with sulfuric acid ( $H_2SO_4$ ) to release fermentable sugars by breaking down hemicellulose and lignin. The fermented mixture underwent distillation, and the resulting bioethanol was analyzed for its physicochemical properties. The bioethanol showed an acid value of 153.7 mg KOH/g, an iodine value of 1.5 ml  $I_2$ /69 mg, a refractive index of 1.34, a density of 1.186 g/ml, a specific gravity of 3.54, and a cloud point of 10°C. FTIR analysis revealed functional groups associated with alcohols, alkenes, ethers, and aromatic compounds, including peaks at ( $3337.90\text{ cm}^{-1}$ ) O-H stretching, ( $1633.00\text{ cm}^{-1}$ ) C=O stretching, ( $1188.38\text{ cm}^{-1}$  and  $1050.48\text{ cm}^{-1}$ ) C-O stretching, and ( $609.30\text{ cm}^{-1}$  and  $579.06\text{ cm}^{-1}$ ) aromatic and structural vibrations. The bioethanol yield was calculated at 60%, indicating moderate efficiency, with some losses during fermentation or distillation. Process optimization, such as refining fermentation time, temperature, or pretreatment methods, could improve the yield. Despite this, the study demonstrates that paper waste is a viable feedstock for renewable bioethanol production, offering a sustainable biofuel resource, particularly in tropical regions.*



## Physicochemical and FTIR Characterization of Cinnamon Oil

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### Abstract

*This study investigates the extraction and characterization of edible oil from cinnamon plant bark. The oil was extracted using Soxhlet extractor with methanol as the solvent at a temperature of 85°C for 6 hours. After extraction, the oil was purified and analyzed for its physicochemical properties, including iodine value 120.5g I<sub>2</sub>/100g, specific gravity 0.838g/cm<sup>3</sup>, and refractive index 1.361. The results showed that the oil has an orange color and a pleasant odor. The iodine value obtained indicates a moderate level of unsaturation while the specific gravity and refractive index confirmed the oil's authenticity and purity. These Suggest the oils potential for use in food and industrial applications. FTIR analysis revealed functional groups such as hydroxyl 3326cm<sup>-1</sup>, carbonyl 1646.4cm<sup>-1</sup>, and C-H stretching of alkanes 2941.8 and 2832.6 cm<sup>-1</sup>, indicating the presence of bioactive compounds. The findings highlight that cinnamon plant bark oil has desirable properties for use as an edible oil with cardiovascular benefits.*



## Synthesis and Characterization of Bioethanol from Bamboo Waste

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### Abstract

*This study investigates the production of bio-ethanol from African bamboo waste, an underutilized agricultural waste. The experiment involved several steps, including sample collection, pretreatment with H<sub>2</sub>SO<sub>4</sub>, fermentation using *Saccharomyces cerevisiae*, and distillation. The physicochemical properties of the produced bioethanol were evaluated, revealing a light brown color, an unpleasant odor, and various chemical characteristics. Key findings include an iodine value of 14 g I<sub>2</sub>/100g, acid value of 2.17 mg KOH/g, and a refractive index of 1.3485. Other properties such as density (1.059 g/ml), specific gravity (0.884g/cm<sup>3</sup>), cloud point (2°C), and a 60% yield were also recorded. Fourier-transform infrared spectroscopy (FTIR) analysis confirmed the presence of functional groups typically found in bio-ethanol, including O-H (3317.22cm<sup>-1</sup>), C=O (1631.40cm<sup>-1</sup>), and C-O (1188.40cm<sup>-1</sup> and 1049.40cm<sup>-1</sup>) stretching. The results suggest that bamboo waste could serve as a viable source for bio-ethanol production, providing an eco-friendly and renewable energy alternative. However, further purification is recommended to improve bio-ethanol quality and meet industry standards.*





## AI-Driven Conversational Systems for Advancing Sustainable Development Goals: A Review

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### Abstract

*Artificial Intelligence (AI) and Machine Learning (ML) are pivotal in driving sustainable development through innovative solutions that enhance accessibility, equity, and efficiency. This paper examines the transformative role of ML-powered chatbots in advancing social good across critical sectors such as healthcare, education, mental health, and public service delivery. By leveraging supervised, unsupervised, and reinforcement learning in conjunction with natural language processing (NLP), these intelligent systems deliver responsive, context-aware interactions that address resource constraints, particularly in underserved communities. The discussion highlights practical implementations, benefits, and emerging trends, including conversational AI, multimodal systems, and integration with IoT, blockchain, and immersive technologies. While these advancements hold immense promise for achieving inclusive and scalable solutions, challenges such as data privacy, algorithmic bias, and ethical governance remain central to sustainable deployment. Ultimately, this work underscores the necessity of responsible AI frameworks to ensure that chatbot innovations contribute meaningfully to the UN Sustainable Development Goals (SDGs).*

**Keywords:** Artificial Intelligence, Machine Learning, Sustainable Development, Chatbots, Ethical AI.



## Production and Quality Evaluation of Swallow Meal Made from Carrot Using Quacker Oats Flour as Binder

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### Abstract

*The study examined the production and evaluation of carrot swallow meal using quacker oat flour as a functional binder. Carrot mash was blended with quacker oat flour at different proportions: A (95:5), B (90:10), C (80:20), D (70:30), and E (60:40). The samples were assessed for proximate composition, mineral content, vitamin A, textural properties, and sensory acceptability. Proximate results showed that moisture (18.08–18.55%), protein (6.82–7.93%), fat (2.33–2.80%), ash (3.19–3.41%), and fibre (4.83–5.21%) increased significantly ( $p < 0.05$ ) with higher oat inclusion, while carbohydrate decreased (64.76–62.12%). Mineral content also improved progressively: calcium (42.60–54.65 mg/100 g), iron (3.14–3.65 mg/100 g), zinc (2.00–2.26 mg/100 g), magnesium (28.50–32.20 mg/100 g), and potassium (210.80–246.75 mg/100 g). Vitamin A declined with reduced carrot proportion, ranging from 646.95 mg/100 g (A) to 541.85 mg/100 g (E). Texture analysis indicated that hardness (5.89–8.53 N), adhesiveness (0.41–0.69 m/J), gumminess (3.62–6.09 N), and chewiness (28.40–52.70 N.mm) increased with oat addition, suggesting firmer, more mouldable swallows. Sensory evaluation revealed high acceptability across samples, with the 80:20 blends (sample C) rated highest in taste (8.90), colour (8.40), appearance (8.30), texture (8.10), mouldability (8.0), and overall acceptability (8.50). The findings demonstrate that oat flour effectively enhances the nutritional, mineral, and functional quality of carrot swallow, while maintaining consumer appeal. The 80:20 formulations offered the best balance of proximate, mineral enrichment, textural integrity, and sensory acceptability, indicating its potential as a functional staple for dietary diversification.*

**Keywords:** Swallow meal, Carrots, Quacker oats, Binders.



## The Relevance of Statistics in Business Studies: Challenges and Solutions

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### Abstract

*This paper examines the relevance of statistics in business studies, highlighting its importance in decision-making and problem-solving. It discusses the challenges faced by students and instructors in teaching and learning statistics, including abstract teaching methods and inadequate quantitative skills. The paper proposes reform-based approaches, such as using real-world examples, technology, and practical applications, to enhance statistical literacy in business students. By exploring the role of statistics in business and industry, this paper aims to contribute to the ongoing discussion on improving statistics education for business students.*



## Microbial Bioremediation Strategies for Environmental Cleanup

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### Abstract

*Environmental pollution caused by industrialization, urbanization, and excessive use of agrochemicals poses serious threats to ecosystems and human health. Conventional remediation techniques such as excavation and landfilling are often costly, hazardous, and unsustainable. Microbial bioremediation has emerged as an eco-friendly, cost-effective, and efficient strategy for the detoxification and restoration of contaminated environments. This report explores microbial bioremediation strategies, focusing on the diverse roles of bacteria, fungi, and algae in degrading pollutants such as hydrocarbons, pesticides, xenobiotics, and heavy metals. Both ex situ and in situ bioremediation techniques—including biopiles, windrows, bioreactors, bioventing, biosparging, and bioslurping—are discussed alongside the enzymatic mechanisms that enhance pollutant degradation. Advances in biotechnology, such as genetically modified microorganisms and microbial consortia, have expanded the application of bioremediation for persistent organic pollutants and recalcitrant compounds. Despite its promise, bioremediation faces limitations such as site-specific constraints, long timeframes, and reduced effectiveness against highly stable contaminants. Nevertheless, with proper optimization—through nutrient supplementation, environmental monitoring, and microbial enhancement—bioremediation offers a sustainable alternative for environmental cleanup. This seminar underscores its growing importance as a global strategy for mitigating environmental pollution while highlighting the need for continued research to address existing challenges and maximize its effectiveness.*

**Keywords:** Bioremediation, Microorganisms, Environmental Pollution, Sustainable Cleanup



## The Assessment of the Microbiological Quality of Sliced Watermelon Fruits Sold at a Market in Offa, Kwara State, Nigeria

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### Abstract

*Watermelon (Citrullus lanatus) is one of the most widely consumed fruits in Nigeria and is often sold in sliced form by street vendors, raising concerns about microbial contamination. This study assessed the microbiological quality of sliced watermelon fruits sold at a Market, Offa, Kwara State, Nigeria. Samples were collected aseptically and subjected to microbiological analysis using serial dilution and spread plate techniques. Isolates were identified through morphological, cultural, and biochemical characterization. The total aerobic plate count ranged from  $7.2 \times 10^6$  to  $9.2 \times 10^7$  CFU/ml, while the total Staphylococcus count ranged from  $1.2 \times 10^6$  to  $8 \times 10^6$  CFU/ml. Coliform counts ranged from  $3.2 \times 10^4$  to  $1.6 \times 10^5$  CFU/ml, and fungal counts from  $3.7 \times 10^5$  to  $1.6 \times 10^6$  SFU/ml. The predominant bacterial isolates were Staphylococcus aureus and Escherichia coli, while fungal isolates included Aspergillus flavus and Aspergillus niger. The microbial loads observed exceeded recommended limits for safe consumption, indicating a potential public health risk. These findings suggest that sliced watermelon fruits could serve as vehicles for foodborne pathogens, thereby posing significant health hazards to consumers. It is recommended that fruit vendors adopt strict hygienic practices, use clean water and utensils, and that public health authorities enforce food safety regulations to minimize contamination risks.*

**Keywords** - Microbiological quality, Watermelon (Citrullus lanatus), Foodborne pathogen, Public health risk.



## Livability and Social Exclusion in African Cities: Focusing on Key Shaping Factors and Intervention Strategies

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### Abstract

*More people now believe that enhancing a city's liveability will improve the quality of life for city dwellers, especially those from developing countries. African cities suffer severe problems with livability and social exclusion, and resolving these problems requires understanding the main determinants and intervention techniques. Previous researchers have viewed social exclusion and liveability separately, however, their connection in Africa is remarkable. Urban liveability and social exclusivity have been severely impacted across Africa due to rapid urbanisation and many economic, social, and environmental concerns. Using narrative systematic review techniques, this study reviewed previous studies, capturing relevant and current trends and emphasising the adverse effects and possible action plans to lessen social isolation and enhance livability in African cities. It underlines the significance of inclusive and sustainable methods to improve urban dwellers' livability. The study found that the recent COVID-19 pandemic, war, insecurity and insurgency across African cities affect social inclusiveness and livability. The current state of African cities requires more than urban planners and policymakers. There is a need for proactive urban governance, alleviating poverty through empowerment, political stability and the will to bring about liveable cities with social inclusiveness in a sustainable manner.*

**Keywords:** Liveability in African cities, Social inclusion, Urbanisation, Social exclusivity.



## Production and Quality Evaluation of Tuwo Meal Made from Maize Flour Binded with Cassava Flour and Cassava Starch

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### Abstract

*This study emphasized on the production and quality evaluation of tuwo meal prepared from maize flour blended with cassava flour and cassava starch. Six formulations were produced: A (100% agric maize), B (80% agric maize + 20% cassava flour), C (80% agric maize + 20% cassava starch), D (100% local maize), E (80% local maize + 20% cassava flour), and F (80% local maize + 20% cassava starch). Functional properties showed significant ( $p < 0.05$ ) variations, with swelling capacity (256.00–286.50%), water absorption capacity (186.00–211.00%), and dispersibility (68.50–76.50%) increasing in cassava-starch blends. Least gelation concentration ranged from 8.25–9.25%, while bulk density values were stable (0.71–0.75 g/ml). Pasting properties revealed that cassava starch blends (C and F) had the highest peak viscosity (3612.50 RVU, 3560.00 RVU), breakdown, and final viscosity, indicating improved thickening ability. Agric maize blends generally recorded lower pasting temperatures (78.05–79.25°C) compared to local maize (80.10°C). Starch composition increased with cassava starch addition (74.60% in C, 74.25% in F), accompanied by reduced amylose and increased amylopectin, enhancing gelation and palatability. Sensory evaluation indicated that all samples were acceptable, but C (8.50) and E (8.50) recorded the highest overall scores, attributed to their desirable texture, mouldability, and appearance. The findings demonstrate that blending maize flour with cassava flour or starch enhances the functional, pasting, and sensory qualities of tuwo meal. In particular, 20% cassava starch or flour incorporation improved viscosity, dispersibility, and consumer acceptability, suggesting their suitability for producing high-quality, functional tuwo meals.*

**Keywords:** Tuwo, maize, cassava flour, cassava starch.





## Preference of Students for Selection of Snacks in Federal Polytechnic Offa

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### Abstract

*This study was carried out to determine the factors contributing to the selection of snacks among students especially for traditional snacks in comparison with modern snacks, consumption patterns, determinants of choice, and perceived nutritional and cultural values. A structured questionnaire was administered to a representative sample of students for data collection. The results revealed that while modern snacks such as biscuits, chocolates, and pastries were widely consumed due to availability, packaging, and convenience, traditional snacks were selected for their affordability, unique taste, and cultural relevance. Over 40% of the students signified preference for both the traditional and modern snacks which highlights the coexistence of both snack types within students' diets. The study concluded that balanced promotion of traditional snacks alongside modern options could enhance dietary diversity, preserve cultural heritage, and stimulate local food industries.*

**Keywords:** *Nigerian snacks, traditional snacks, modern snacks, student preference, consumption pattern.*



## Challenges of Cybersecurity in Nigeria's Financial Sector

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### Abstract

*Cybersecurity has emerged as a critical pillar of contemporary information technology, with the protection of sensitive data ranking among the foremost challenges of the digital age. The growing reliance on online platforms has significantly increased exposure to threats such as phishing, insider breaches, ransomware, and cryptocurrency-driven cybercrimes. This study explores the cybersecurity challenges confronting Nigeria's financial sector, highlighting key vulnerabilities and proposing strategic interventions. Drawing from recent literature and empirical evidence, it emphasizes the influence of weak regulatory frameworks, limited awareness, and the dynamic evolution of cyber threats. The findings underscore the urgent need for strengthened policies, advanced encryption systems, and the cultivation of a robust cybersecurity culture across financial institutions. Furthermore, the study evaluates the existing legal frameworks governing cybersecurity in the sector, assessing their effectiveness and limitations. Ultimately, it outlines major challenges and presents context-specific recommendations aimed at enhancing cybersecurity resilience within Nigeria's financial sector.*

**Keywords:** Cybersecurity, Cryptocurrency-Driven, Cyber Threats, Phishing, Ransomware, Cybersecurity Challenges.



## Examining the Effect of Artificial Intelligence Usage on the Fourth Estate of the Realm in Nigeria

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### Abstract

*Adopting and integrating artificial intelligence (AI) into contemporary practice of journalism has brought a far-reaching and unimaginable implications in the contemporary society. This paper leaning on qualitative review of literature examines the effect of artificial intelligence usage on the fourth estate of the realm in Nigeria. This paper adopts the Technology acceptance model (TAM) as its theoretical framework specifying data-driven reporting, automated journalism and personalised news reporting as some of the obvious results of artificial intelligence usage in the fourth estate of the realm which have invariably brought a revolution to journalism practice of news generation and dissemination. These developments impact field work and newsroom thereby fostering a high-paced content creation and enhanced insights into data analysis. Nonetheless, AI technologies equally pose some challenges for the media industry including misinformation and disinformation through algorithmic biases, potential job losses and ethical concerns that border on privacy invasion. This paper while highlighting some of these challenges and opportunities inherent in AI use also recommends the need to balance AI's benefits with high ethical standards to maintain the media's penchant for objectivity, fairness and integrity.*

**Keywords:** artificial intelligence, media practice, journalism, media ethics, AI, machine learning



## Forecasting Petroleum Motor Spirit (PMS) Prices in Nigeria Using Time Series Analysis: ARIMA and SARIMA Approaches

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### Abstract

*Petroleum Motor Spirit (PMS) pricing is a critical determinant of Nigeria's economic stability, influencing transportation, production costs, and inflation. Despite being a major oil producer, Nigeria faces persistent PMS price volatility due to global oil market fluctuations, exchange rate instability, and inefficiencies in the downstream sector. The 2020 removal of subsidies marked a structural shift, triggering sharp increases in PMS prices and widespread socio-economic impacts. This study applies time series analysis to model and forecast PMS prices in Nigeria using monthly data from 2015 to 2023, obtained from the National Bureau of Statistics (NBS) and the Nigerian National Petroleum Corporation (NNPC). Descriptive analysis, stationarity testing with the Augmented Dickey-Fuller (ADF) test, and model estimation using ARIMA and SARIMA frameworks were conducted. Model adequacy was assessed using AIC, BIC, and RMSE, with residual diagnostics to validate robustness. Results show that PMS prices remained relatively stable under subsidy regimes but surged significantly following deregulation, reaching over ₦670 per litre in 2023. The ARIMA(5,1,1) model provided the best fit, yielding the lowest RMSE (31.28) and reliable short-term forecasts. Forecasts for 2024 indicate a steady upward trend, with prices projected to exceed ₦780 by year-end, alongside increasing uncertainty. The findings highlight the strong link between PMS pricing, inflationary pressures, and household welfare. This study concludes that robust statistical forecasting can support evidence-based policymaking. It recommends controlled deregulation, expansion of domestic refining, targeted subsidies for vulnerable groups, and adoption of data-driven strategies to stabilize PMS pricing in Nigeria.*



## AI-Driven Cybersecurity: A Federated, Privacy-Preserving and Explainable Framework for Real-Time Threat Detection

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### Abstract

*Cyber-attacks are increasing in both frequency and sophistication, creating major challenges for organisations seeking to protect their digital assets. This paper presents AI-SIEM, an artificial-intelligence-driven cybersecurity framework that is federated, privacy-preserving, and explainable. Unlike traditional, reactive defences, AI-SIEM integrates deep-learning models, fully connected, convolutional, and recurrent neural networks with advanced feature engineering to detect anomalies, prioritise alerts, and execute automated responses in real time. To evaluate the framework, benchmark intrusion-detection datasets and simulated enterprise traffic were used, and performance was measured on detection accuracy, false-positive rate, and detection latency. The ensemble model achieved 98.3 % detection accuracy, 1.7 % false positives, and an average latency of 74 ms. Therefore, outperforming signature-based systems and recent AI-driven approaches. The paper also highlights emerging trends such as privacy-preserving machine learning, behavioural analytics, and adaptive zero-trust architectures as enablers of the next generation of resilient digital security.*

**Keywords:** Artificial Intelligence, Cybersecurity, Real-Time Threat Detection, Federated Learning, Privacy-Preserving Machine Learning, Explainable AI.



## A Bibliometric Analysis of Artificial intelligence Using Machine learning in Sustainable Development

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### Abstract

*The integration of artificial intelligence (AI), machine learning (ML), and Internet of Things (IoT) technologies is reshaping diverse sectors including mining, agriculture, energy management, language translation, and sustainability-focused systems. Recent scholarly contributions highlight the rapid expansion of research on AI-driven optimization, predictive modelling, and intelligent decision-making across industrial and academic domains. For instance, studies have explored machine learning-based powder factor optimization for sustainable mining, AI-enabled translation pedagogy in higher education, IoT-driven crop prediction frameworks, and explainable AI for agricultural yield estimation. However, there are few studies on the bibliometric analysis of these topics. Therefore, the aim of this study is to explore bibliometric analysis method using Scopus database as the source of data set. This bibliometric review synthesizes trends from recent literature spanning 2020–2026, mapping the interdisciplinary trajectory of AI research towards sustainability, precision agriculture, intelligent manufacturing, and ethical system design. The findings reveal the source, the countries, the authors that have been contributing to the topic and where there is the need to fill in the gap such as African countries that have little or no contribution. By so doing, they can contribute to understanding the intellectual structure of this emerging knowledge base and identify gaps for future research, particularly in integrating explainability, cross-domain transferability, and socio-ethical implications into AI-driven applications.*

**Keywords:** Artificial intelligence, Sustainability, Bibliometric Analysis, Machine Learning, Development.



## Programming: A Golden Ticket to Job Security

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### Abstract

*In today's technology-driven economy, programming languages have become a pivotal determinant of employability, adaptability, and long-term job security across industries. As organizations accelerate digital transformation, proficiency in languages such as Python, Java, C++, and emerging domain-specific languages is no longer optional but a core prerequisite for sustained career growth. This paper explores the connection between programming language competence and job security, drawing from recent industry surveys, labour market analyses, and academic studies. It highlights how programming languages not only serve as technical tools but also as problem-solving tools, innovation, and professional resilience in volatile job markets. Furthermore, the paper investigates trends in language demand, employer preferences, and the rise of polyglot programmers who leverage multiple languages to remain competitive. Findings suggest that continuous learning, coupled with strategic mastery of in-demand programming languages, significantly enhances job security by ensuring alignment with evolving technological ecosystems. This research contributes to both academia and practice by providing evidence-based insights into the role of programming language expertise as a golden ticket for sustainable employment in the digital age.*

**Keywords:** Programming languages, Job security, Employability, Digital skills, Workforce adaptability.





## Microbial Contamination of Restaurant Spoon and Dish Washing Water of a Tertiary Institution in Offa, Kwara State

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### Abstract

*The cafeteria (also known as restaurant) is a place where foods are prepared and sold. It is a means of livelihood for a significant number of people in developing countries but the poor hygiene conditions in some of these cafeterias raise worries on microbiological safety. This study was designed to isolate and identify microbial contaminants in spoon and dish washing water of a tertiary institution restaurant, Offa, Kwara State. The dish washing water and spoons were collected from two (2) tertiary institution restaurants. Pour plate and Streak plate method were used in this analysis. The organisms were identified using standard conventional methods which include macroscopic, microscopic and biochemical characterization. The total aerobic plate count ranged from  $20.0 \times 10^4$  to too numerous to count, while the total fungal and total coliform count were too numerous to count. The bacterial species identified were ***Pseudomonas aeruginosa***, ***Bacillus sp***, and ***Staphylococcus aureus***. The fungal species identified were ***Aspergillus sp***, ***Rhizopus sp***, and ***Penicillium sp***. The study revealed critical hygiene concerns in the analyzed spoon and dish washing water, identifying bacterial and fungal contaminants that pose significant health risks.*

**Keywords:** Isolation, Identification, Microbial contamination, Food contact surfaces.



## Modeling Consumer Price Index with Autoregressive Integrated Moving Average in Nigeria

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### Abstract

*Consumer Price Index (CPI) is an important indicator used to measure inflation. The main objective of this research was to select the best univariate time series model that can forecast Monthly Consumer Price Index in Nigeria. Consumer Price Index data were collected from January 2000 to December 2020. Box and Jenkins methodology of analyzing time series data, which is identification of parameter, estimation of parameter and model diagnoses was adopted. The best model was identified and estimated by using Correlogram (Autocorrelation Function and Partial Autocorrelation function) and diagnosed by Ljung-Box test, Correlogram to check serial correlation in the residuals and significance spikes in lags respectively. The best model, Autoregressive Integrated Moving Average ARIMA (1,1,1) with Root Mean Square Error (RMSE) of 1.1139 and Akaike criteria (AIC) of 772.43 was selected. Lung-Box test ( $P$ -value= 0.6952 and  $p > .05$ ) also validate the model. The model was selected as the best for Consumer Price Index (CPI) in Nigeria since it showed smallest errors measures. Lastly, a forecast was done using the ARIMA (1,1,1) model. Forecasting CPI is an important factor for any economy because it is essential in economic planning for the future. Hence, identifying a more accurate forecasting model is a major contribution to the development of Nigeria.*

**Keywords:** Consumer price index, correlogram, Arima and Model.



## Application of TGARCH on Exchange Rate, Systems and Performance of Nigeria Foreign Exchange Market

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### Abstract

*Foreign Exchange Market on its own is a volatile market because of the unstable nature of exchange rate due to many factors which exchange rate policy is one of them. The volatility of exchange rate and the measure of volatility has been of great concern to investors and foreign exchange operators. In Nigeria the four foreign currencies that are mostly traded upon are the US Dollars (USD), the Great Britain Pound (GBP), the EURO and CFA which the data was collected from June 2016 to February 2025 for this study. This study assessed the returns/volatility of exchange rate and the impact cause on it by two different regimes of exchange rate in regulating the activities of Foreign Exchange (Forex) Market in Nigeria. The AR+TGARCH was adopted because TGARCH accounts for asymmetry in volatility (leverage effect) and allows the conditional variance to depend on the sign of lagged innovation. The results from the analysis showed that all factors that influence returns and volatility including exchange rate system were significance in at least in one of the series. The previous returns was the factor that affects current returns mostly, because it was significant in three of the four financial time series. While the previous shocks and previous volatility were significant in all four series. Based on these submission, the previous returns, previous shocks and previous volatility are the major factors to be considered in exchange rate returns and volatility of exchange rate.*

**Keywords:** Exchange Rate, Volatility, Performance, Returns, Shocks



## Exchange Rate, Stock Market Volatility and Performance: An Assessment

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### Abstract

*The study of financial time series has been of important to guide investors, because of its volatility. This led to the investigation of the effect of exchange rate on the stock price returns and volatility. The two financial time series in this study are Skye Shelter Stock price and Exchange Rate of Nigerian Naira (NGN) on American Dollar (USD) from 2019 to 2024 and the data was collected from reliable sources. The model applied on the time series data was  $AR(1)+TGARCH(1,1)$  with assumption of student's  $t$  distribution, because most financial data are fat tail. The R Statistical Software version 4.4.3 was used in analysing the data. The estimated parameter of the exchange rate did not have significant effect on the returns, but there was significant effect on volatility. The exchange rate decreases return, while it increases volatility.*

**Keywords:** Exchange rate, Stock Price, Volatility, Returns, Performance.

